

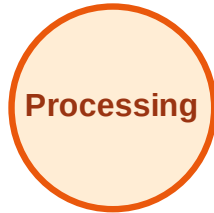
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



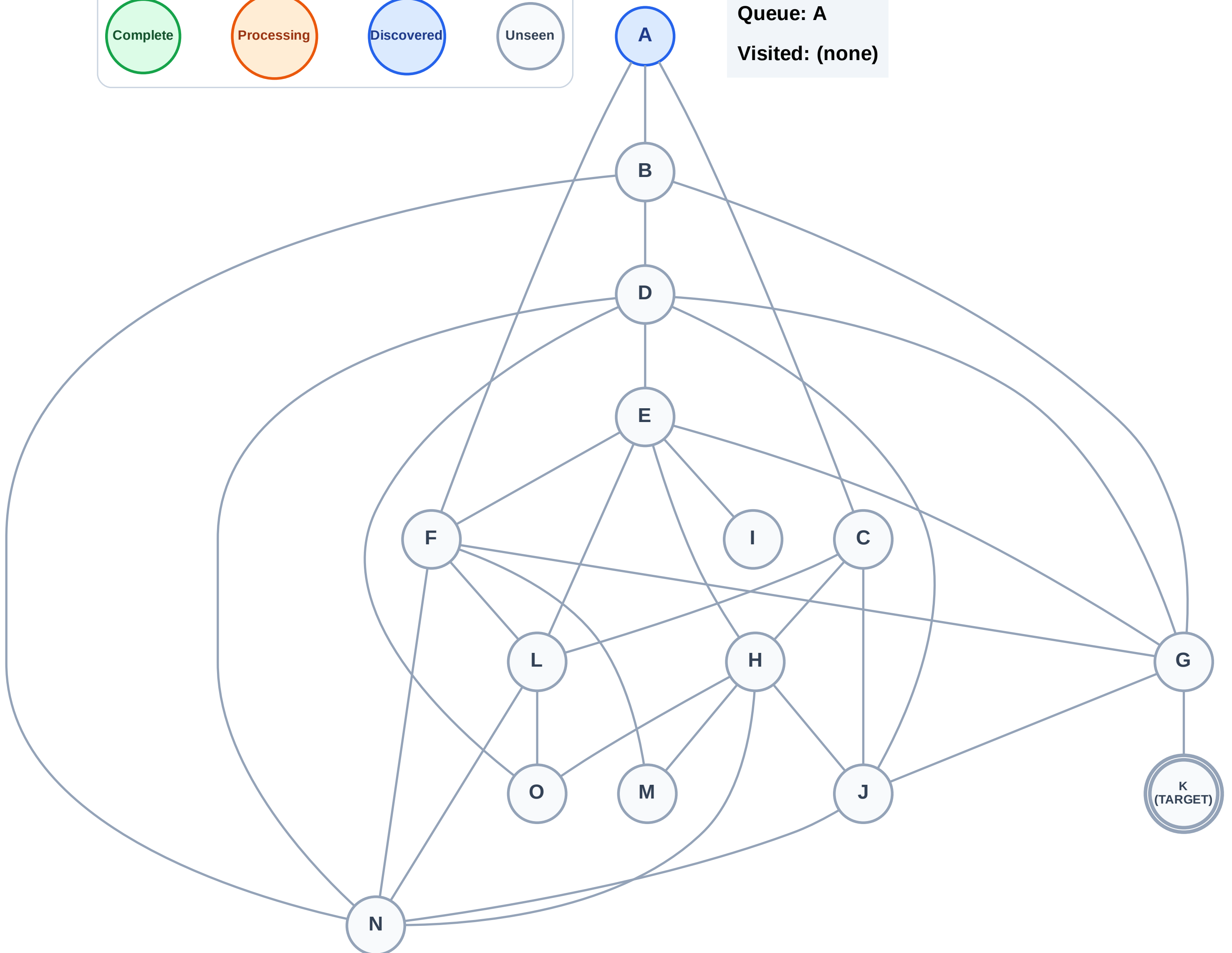
Discovered



Unseen

Queue: A

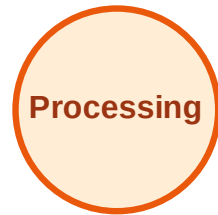
Visited: (none)



BFS Traversal

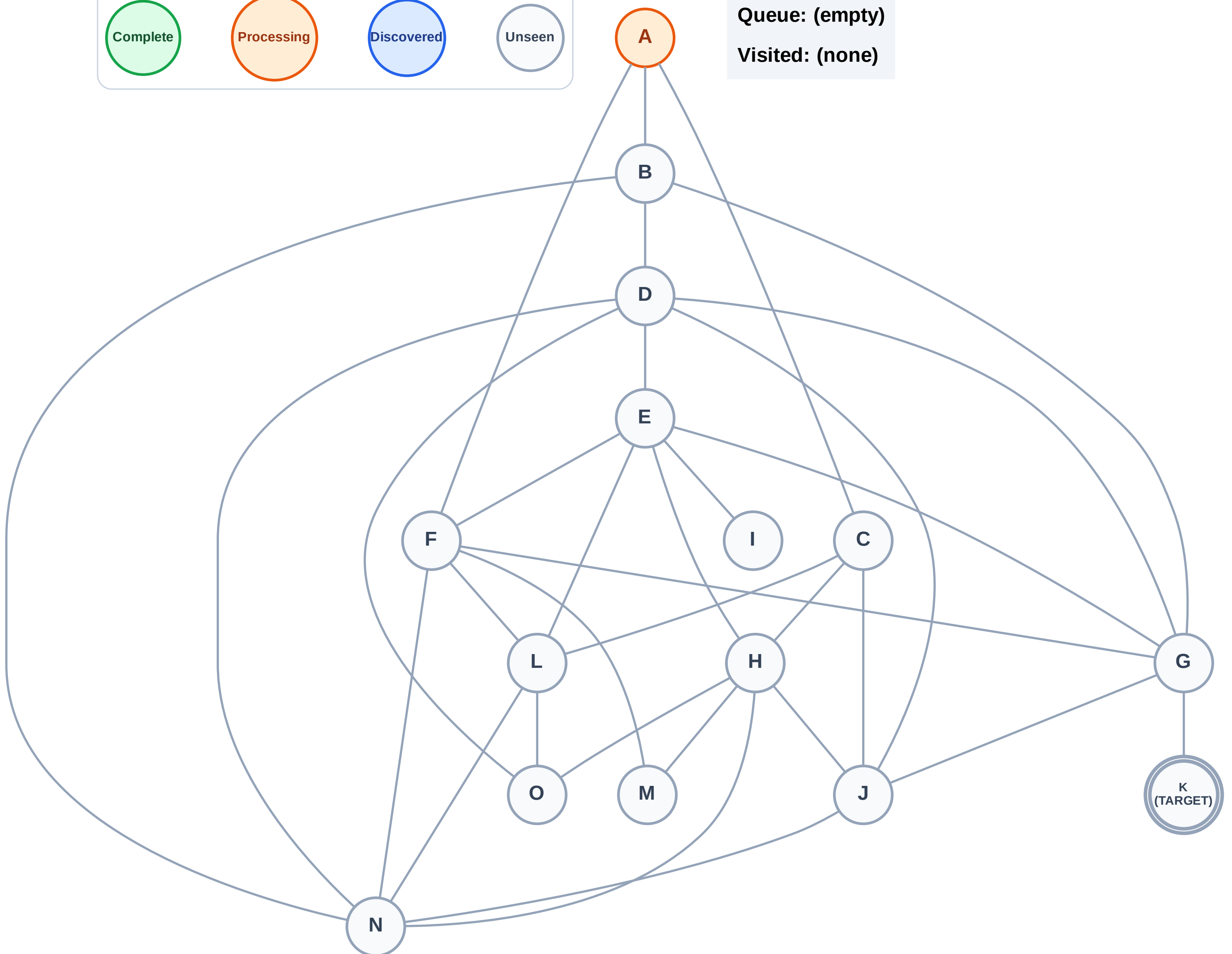
Step 2: Process node A

Legend



Queue: (empty)

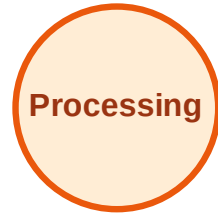
Visited: (none)



BFS Traversal

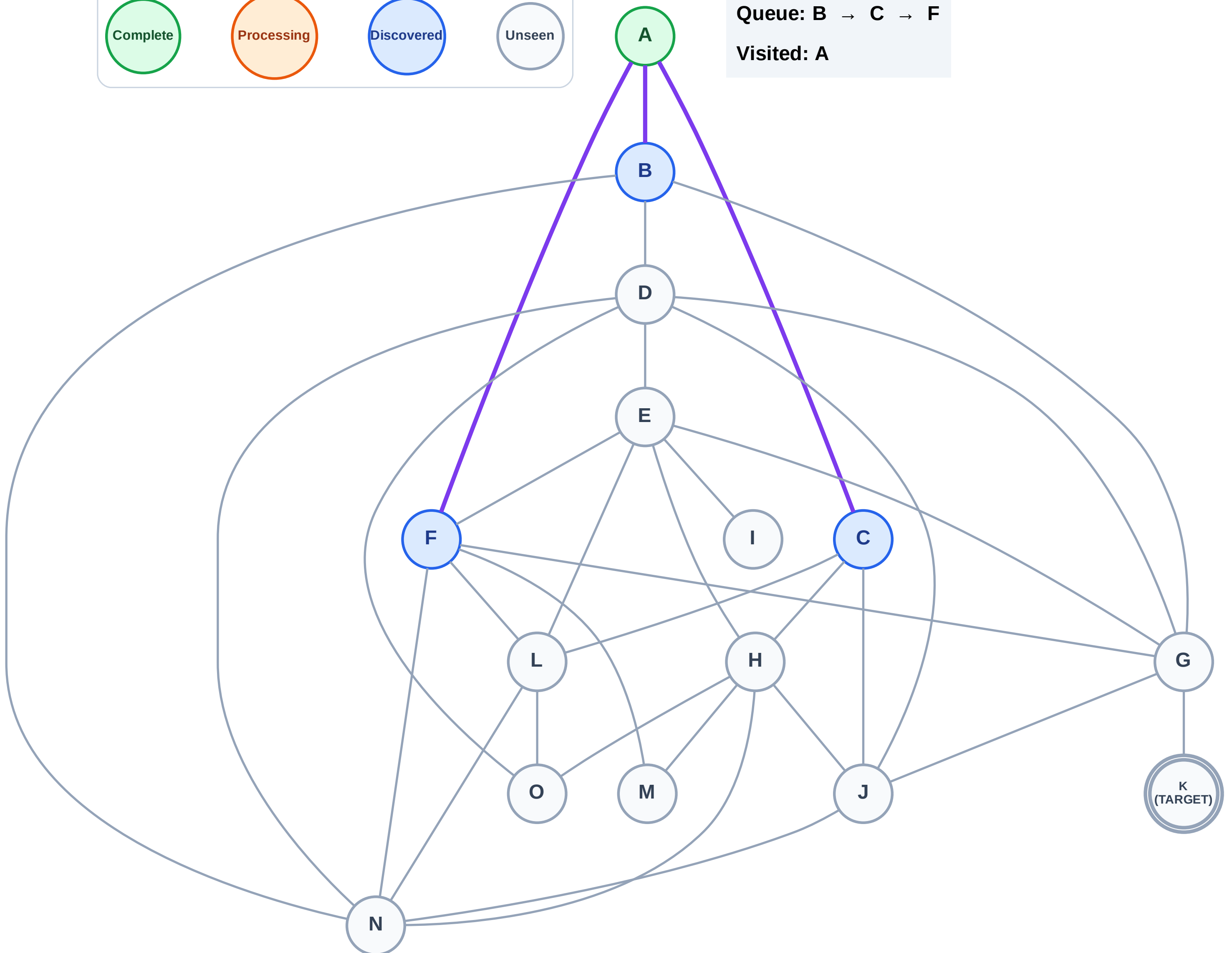
Step 3: Discover B, C, F from A

Legend



Queue: B → C → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

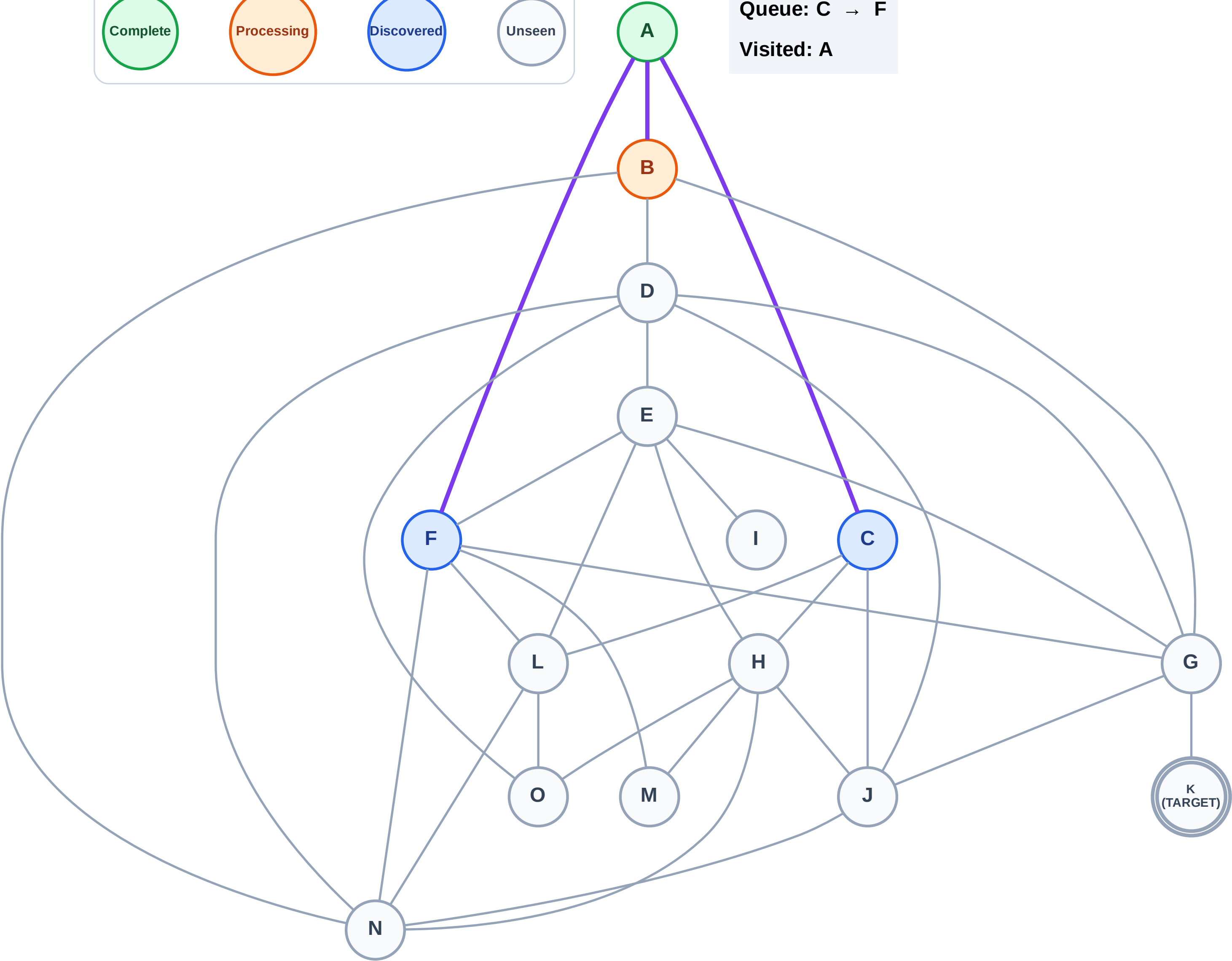
Processing

Discovered

Unseen

Queue: C → F

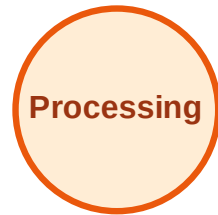
Visited: A



BFS Traversal

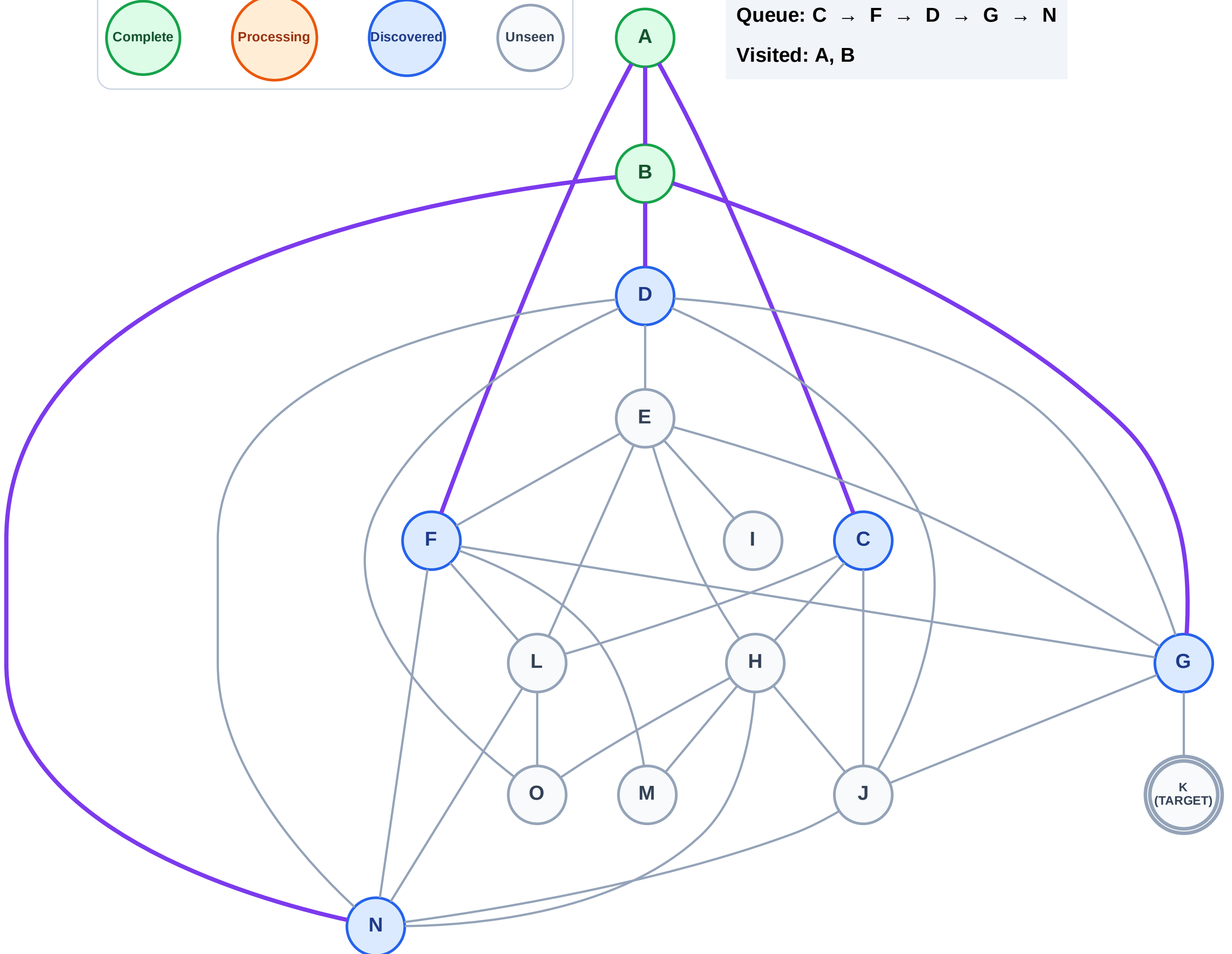
Step 5: Discover D, G, N from B

Legend



Queue: C → F → D → G → N

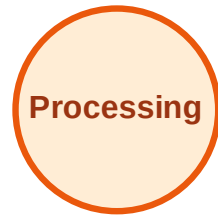
Visited: A, B



BFS Traversal

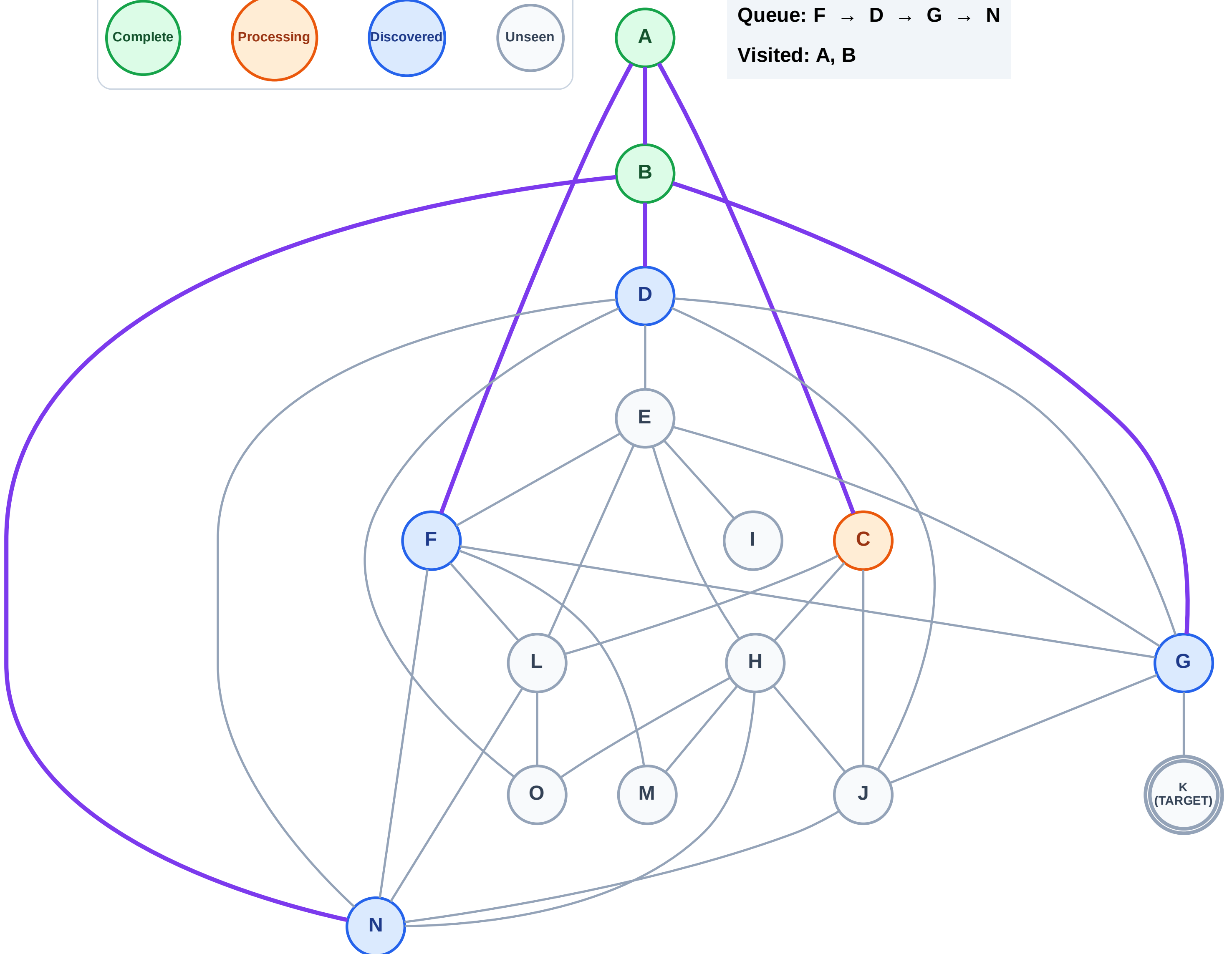
Step 6: Process node C

Legend



Queue: F → D → G → N

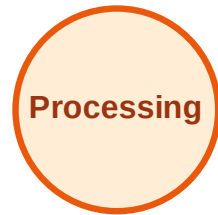
Visited: A, B



BFS Traversal

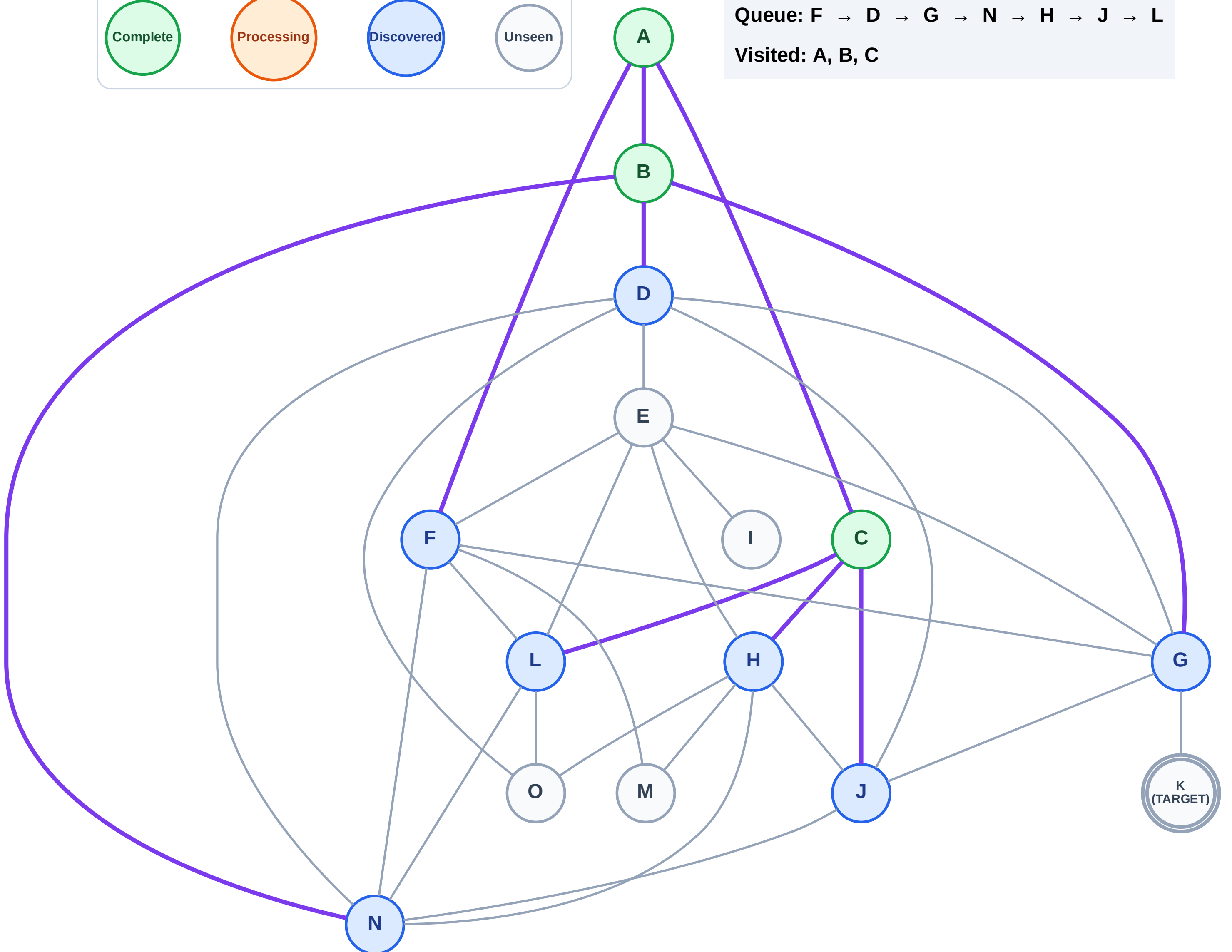
Step 7: Discover H, J, L from C

Legend



Queue: F → D → G → N → H → J → L

Visited: A, B, C



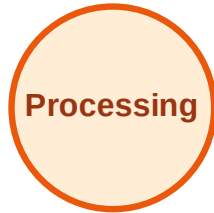
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



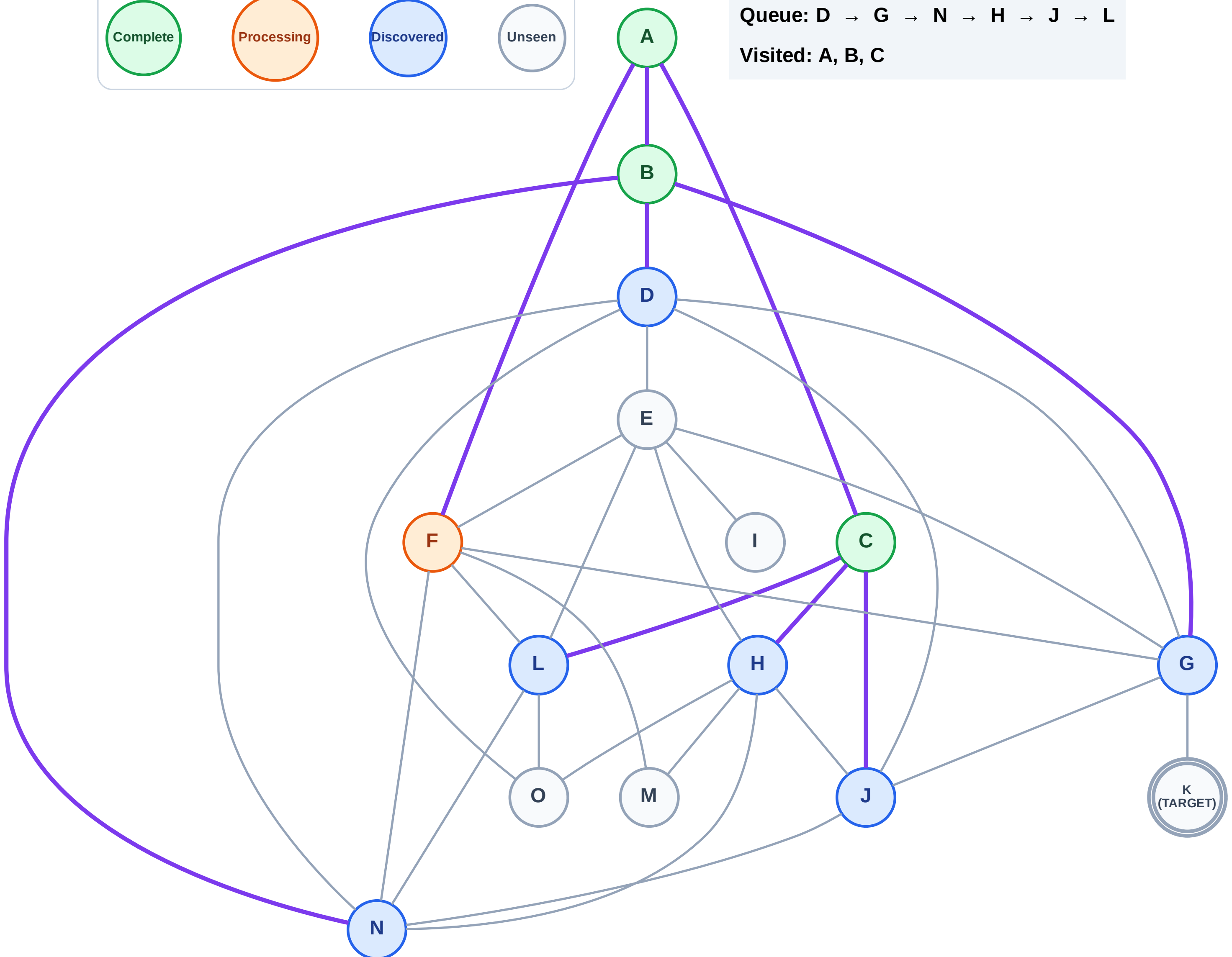
Discovered



Unseen

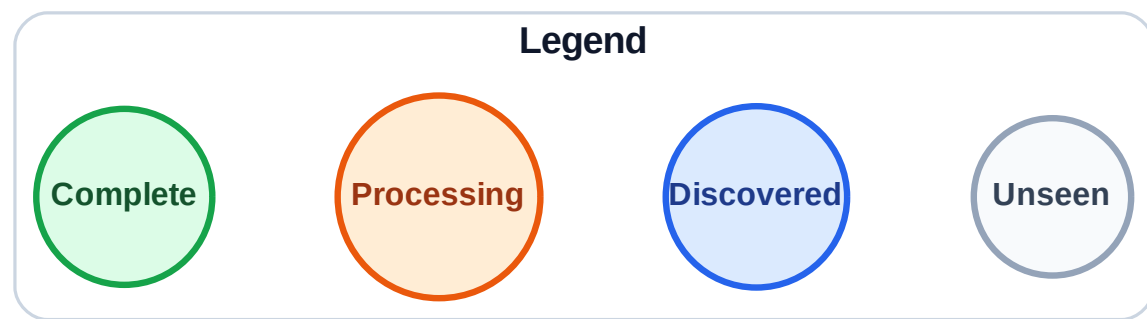
Queue: D → G → N → H → J → L

Visited: A, B, C



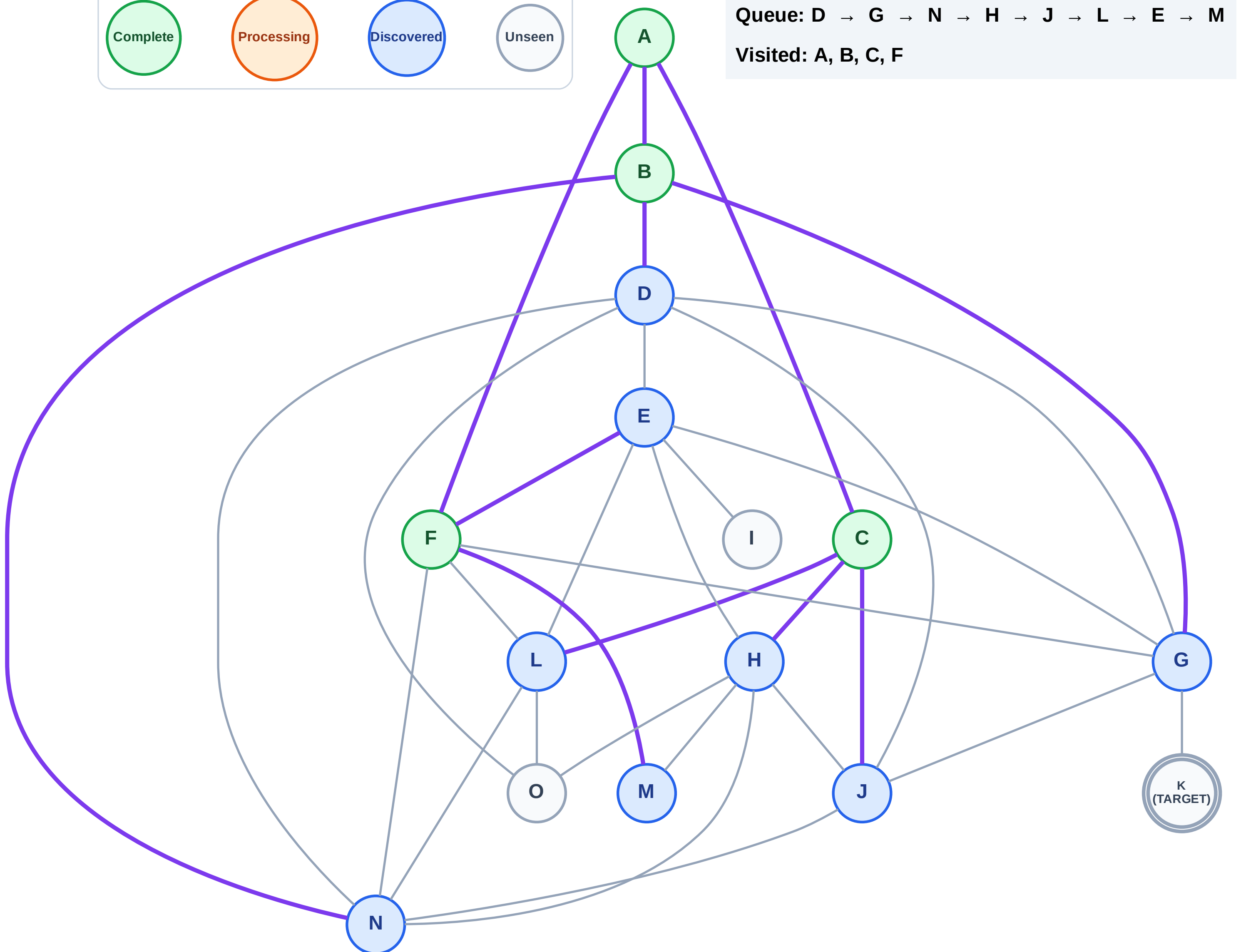
BFS Traversal

Step 9: Discover E, M from F



Queue: D → G → N → H → J → L → E → M

Visited: A, B, C, F



BFS Traversal

Step 10: Process node D

Legend

Complete

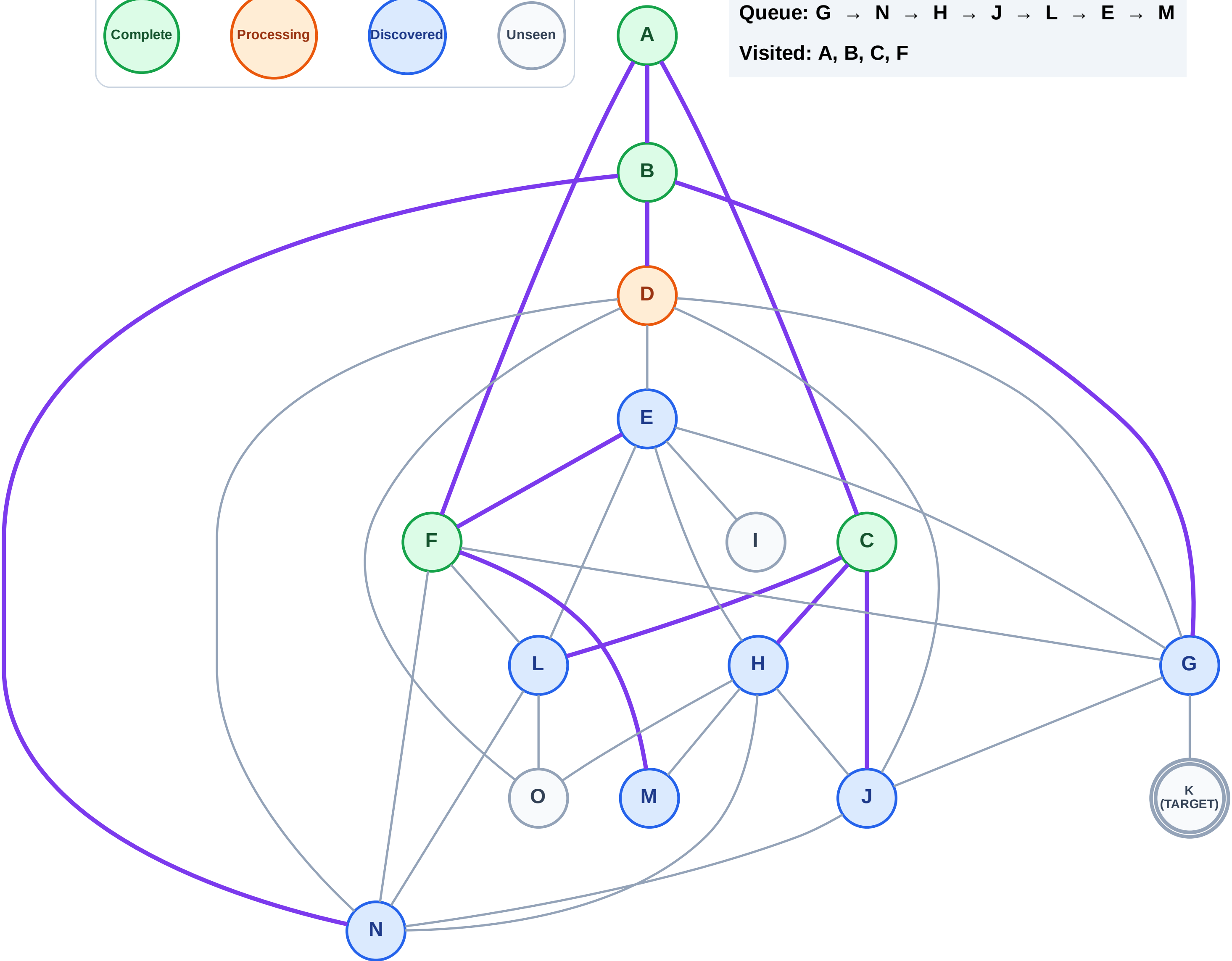
Processing

Discovered

Unseen

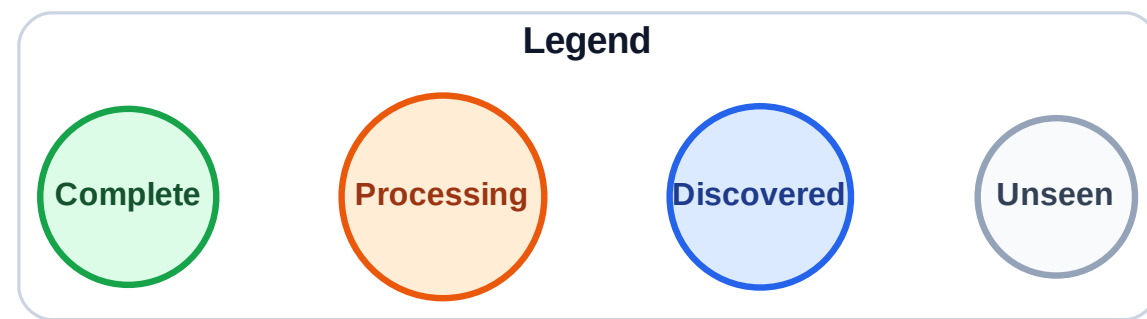
Queue: G → N → H → J → L → E → M

Visited: A, B, C, F



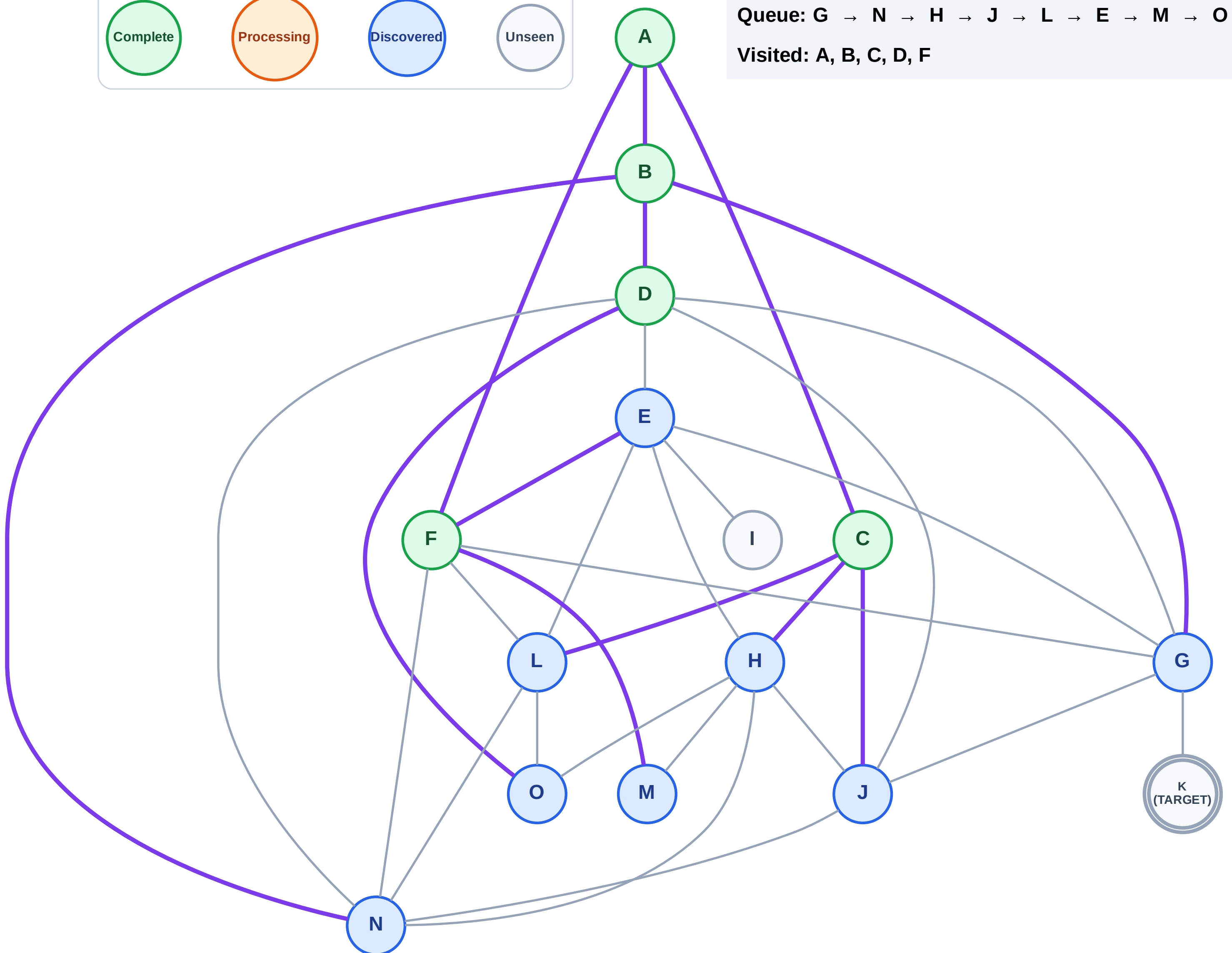
BFS Traversal

Step 11: Discover O from D



Queue: G → N → H → J → L → E → M → O

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node G

Legend

Complete

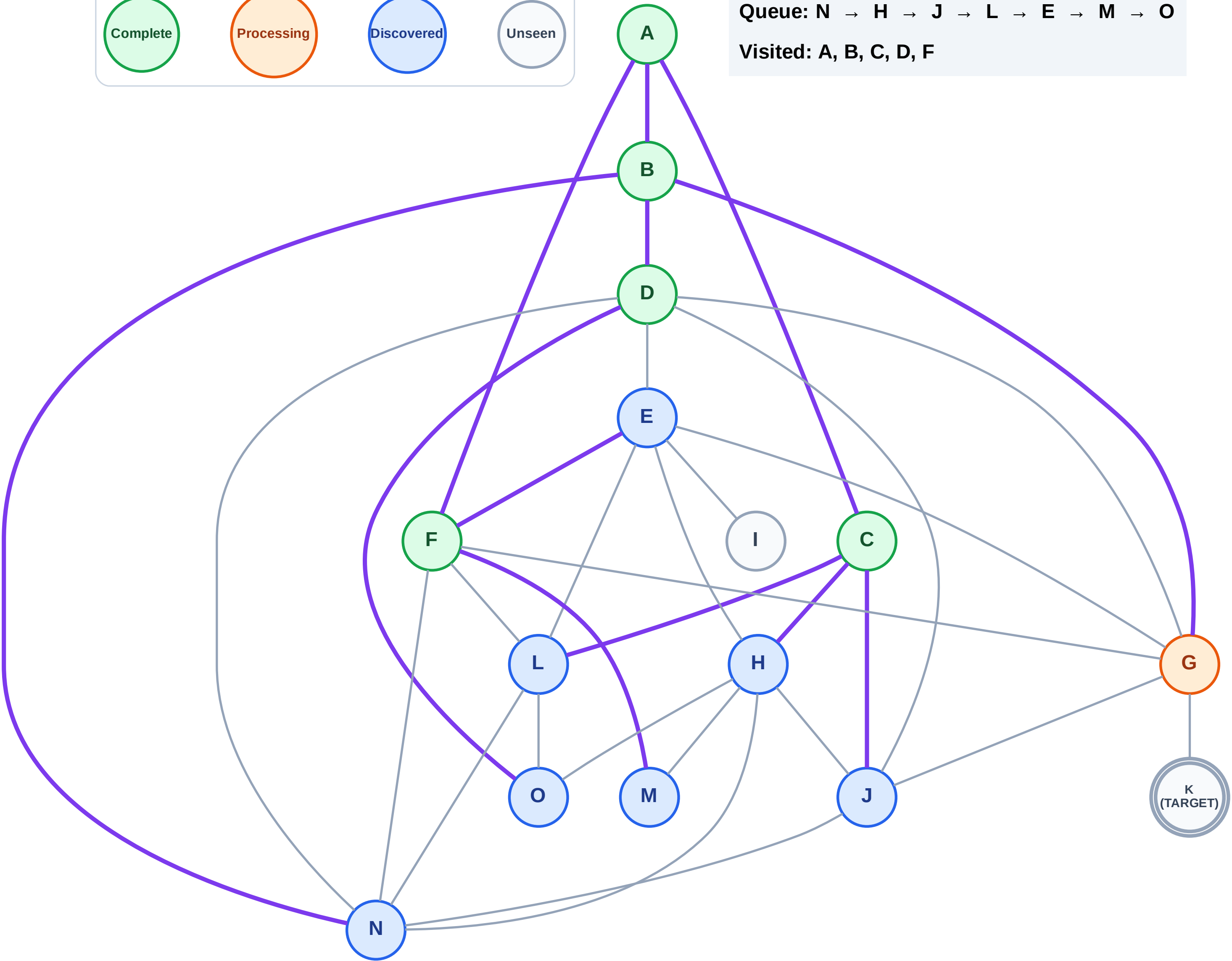
Processing

Discovered

Unseen

Queue: N → H → J → L → E → M → O

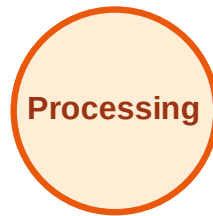
Visited: A, B, C, D, F



BFS Traversal

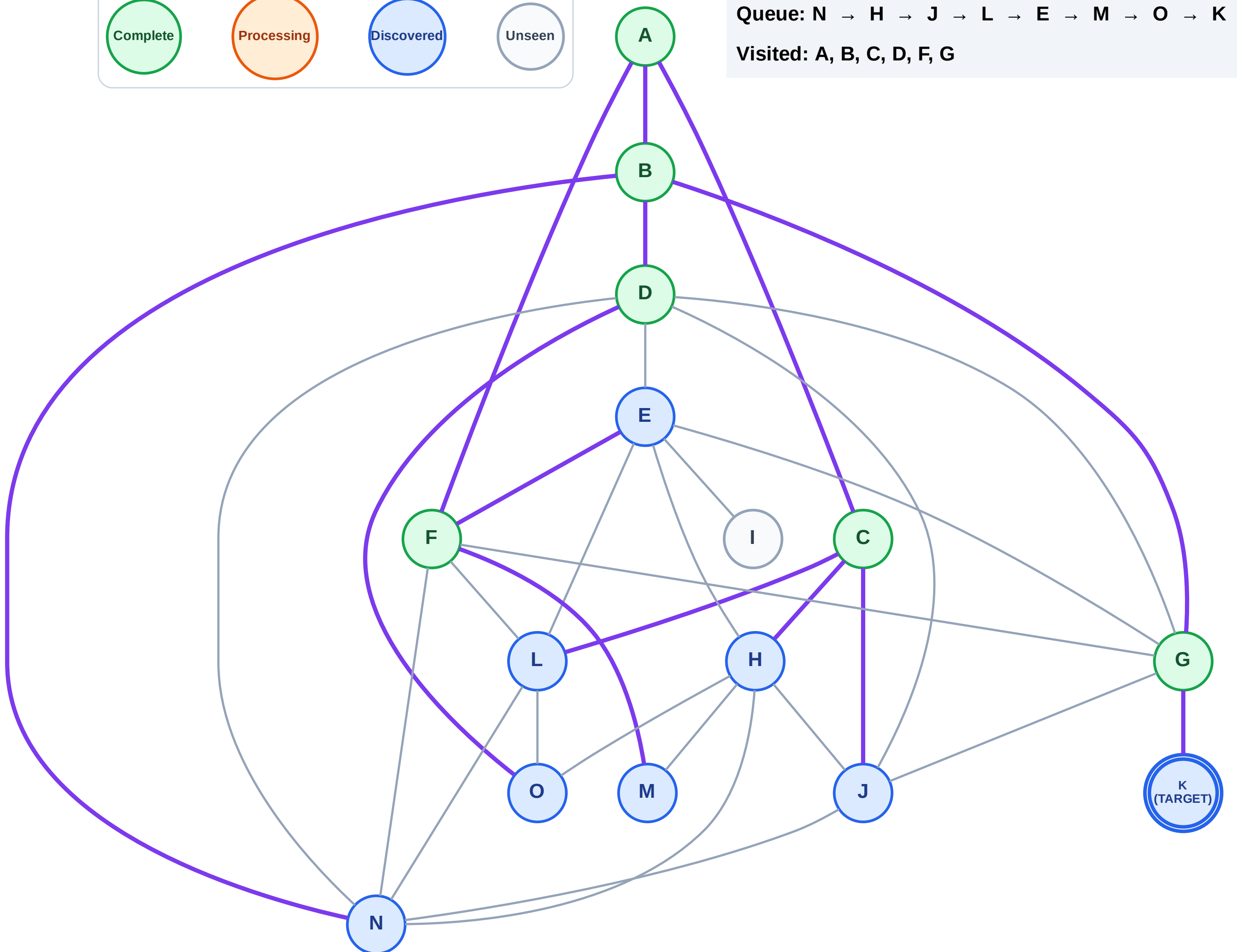
Step 13: Discover K from G

Legend



Queue: N → H → J → L → E → M → O → K

Visited: A, B, C, D, F, G



BFS Traversal

Step 14: Process node N

Legend

Complete

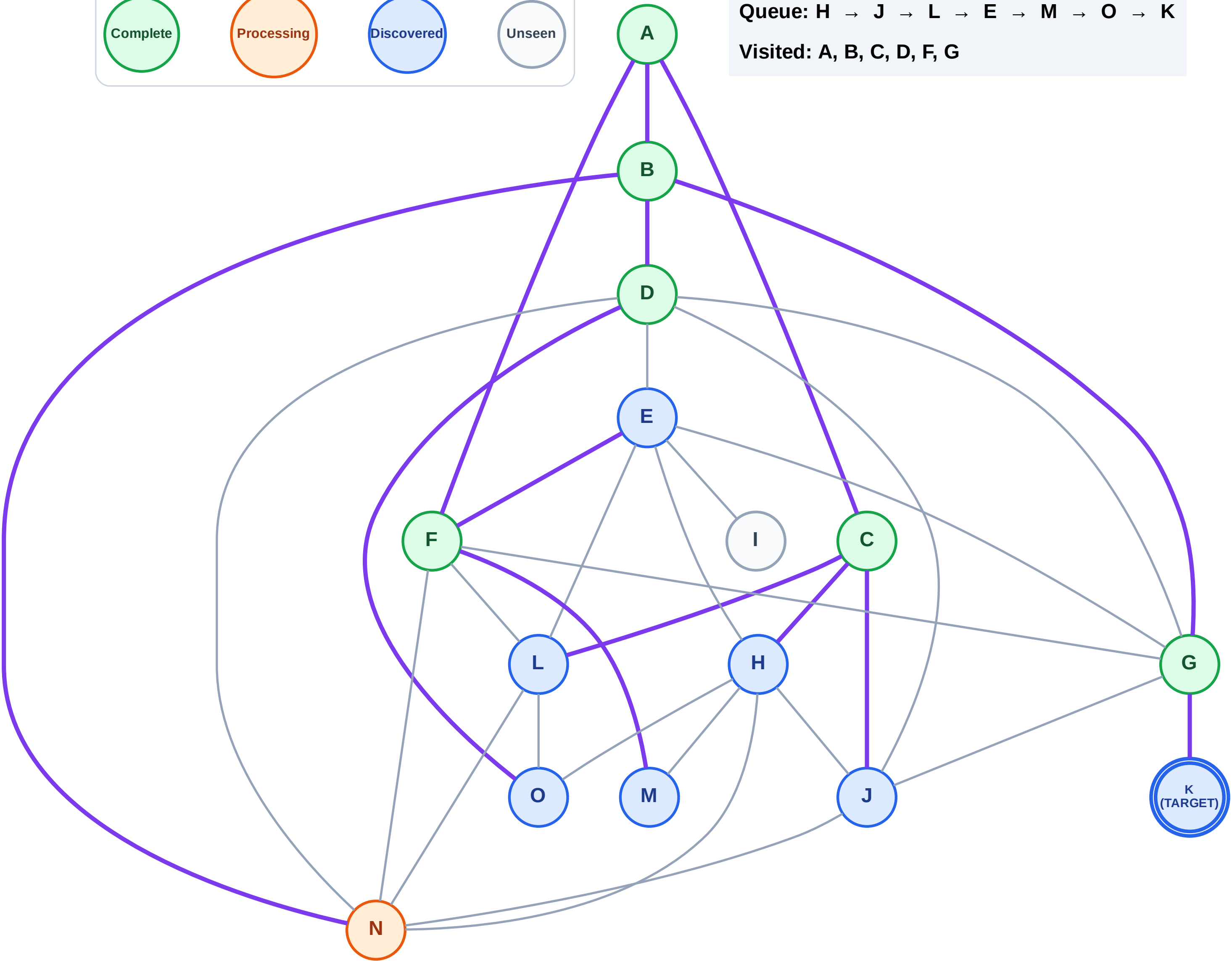
Processing

Discovered

Unseen

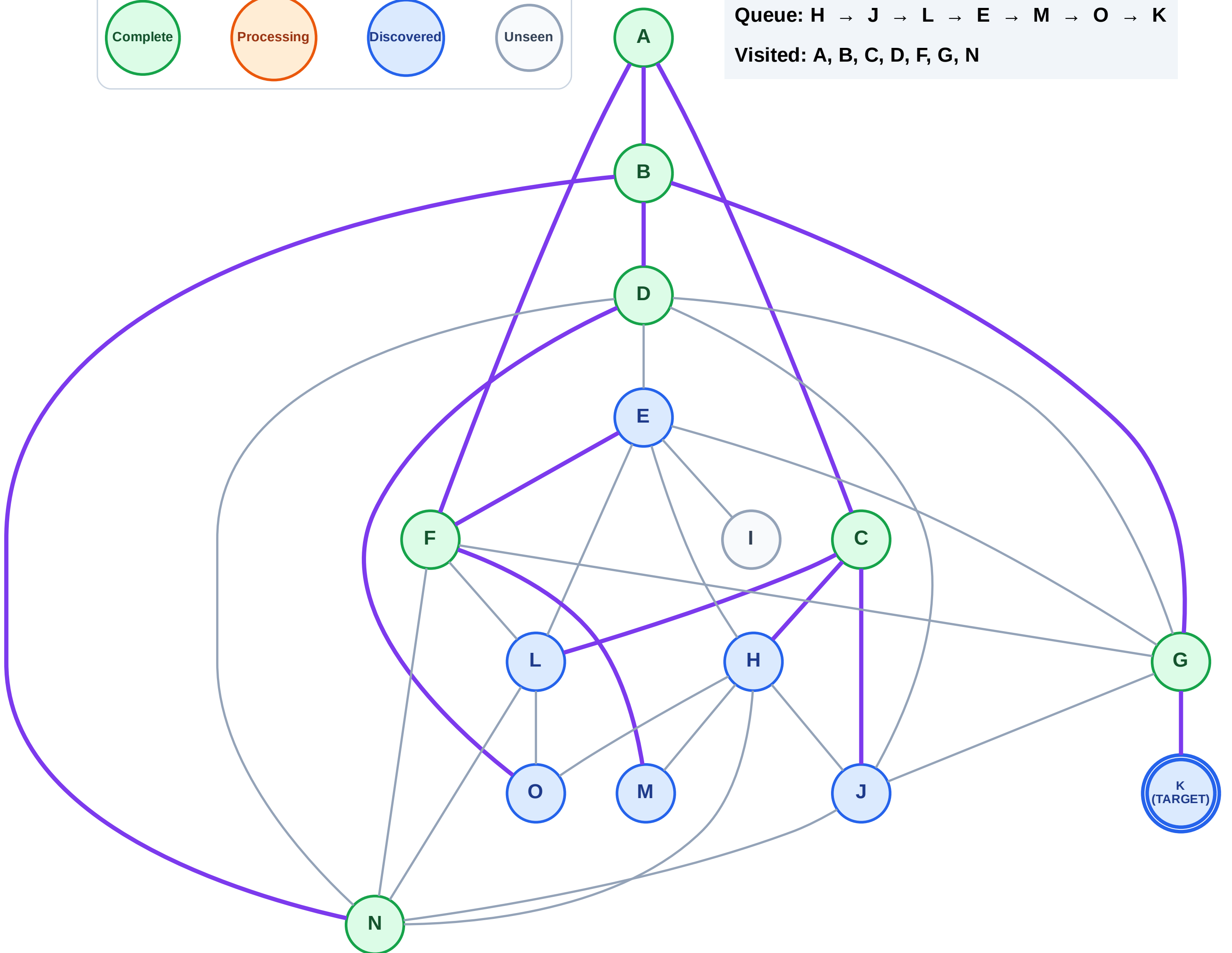
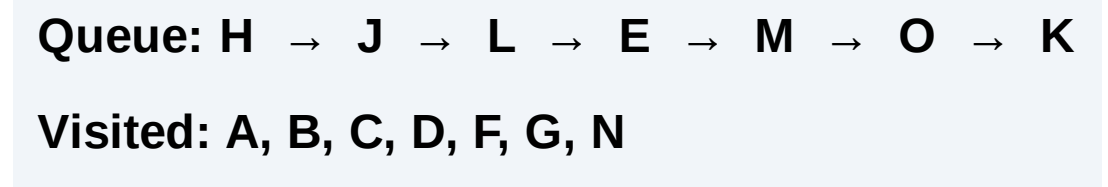
Queue: H → J → L → E → M → O → K

Visited: A, B, C, D, F, G



Step 15: No new neighbors from N

Step 15: No new neighbors from N



BFS Traversal

Step 16: Process node H

Legend

Complete

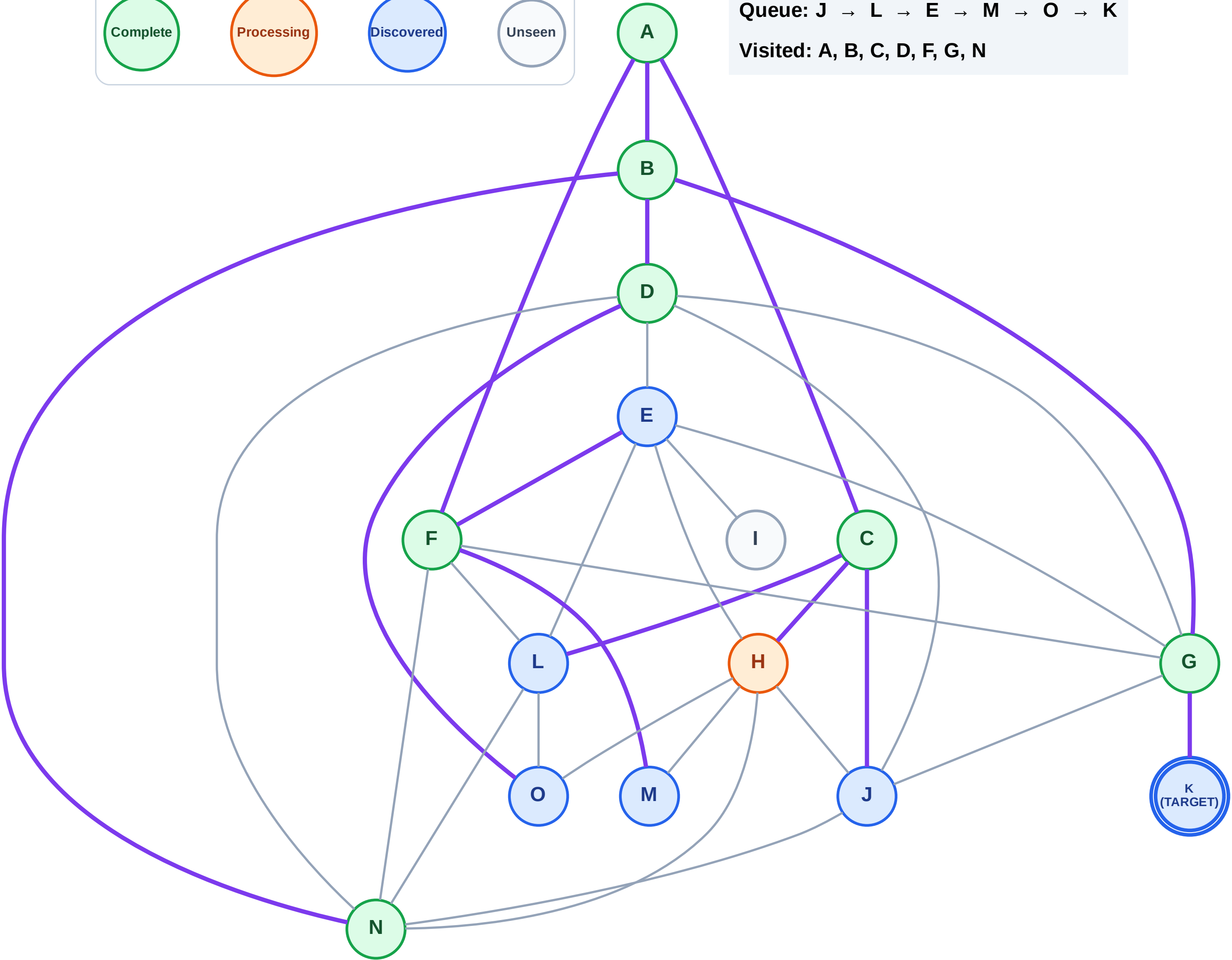
Processing

Discovered

Unseen

Queue: J → L → E → M → O → K

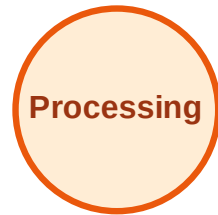
Visited: A, B, C, D, F, G, N



BFS Traversal

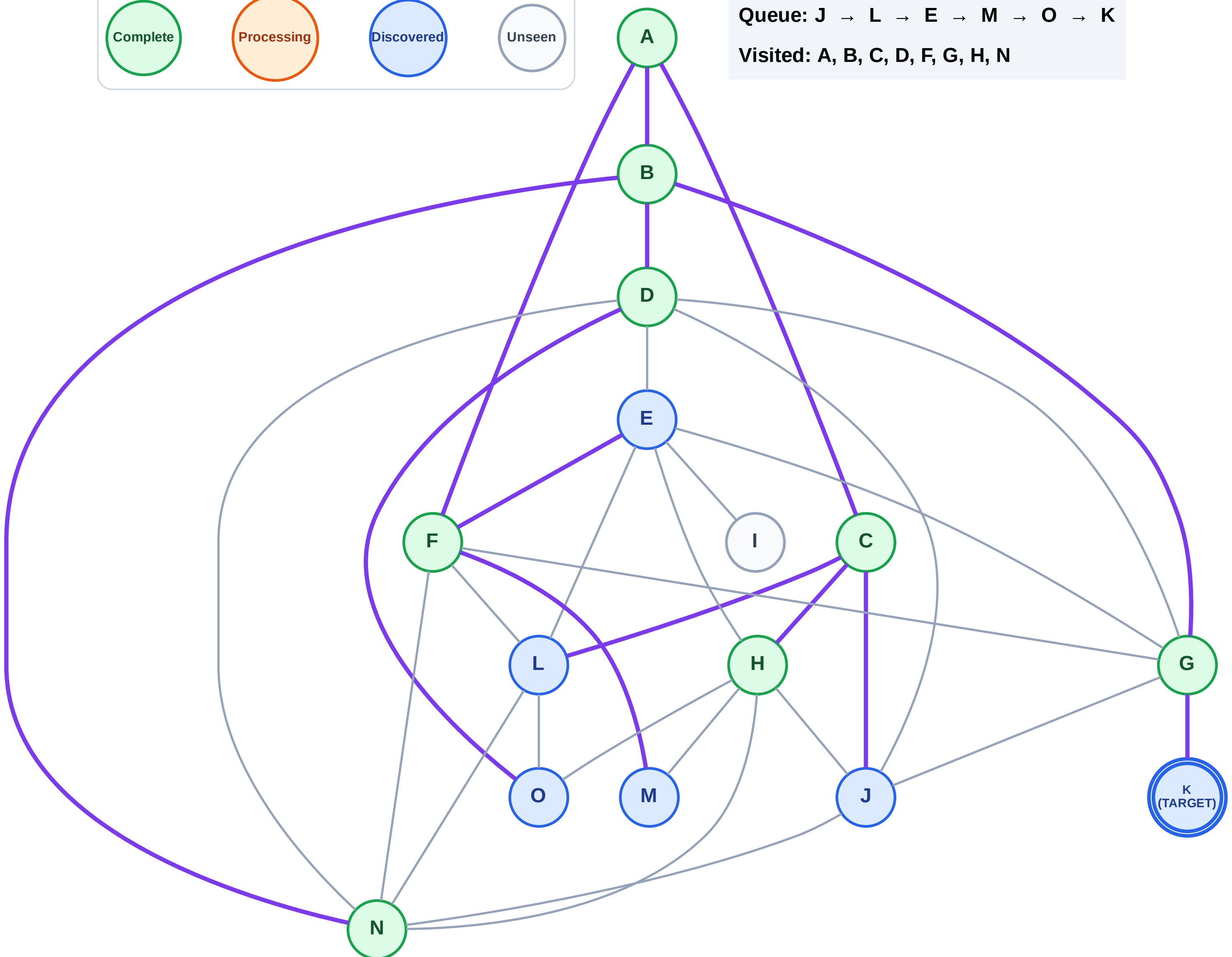
Step 17: No new neighbors from H

Legend



Queue: J → L → E → M → O → K

Visited: A, B, C, D, F, G, H, N



BFS Traversal

Step 18: Process node J

Legend

Complete

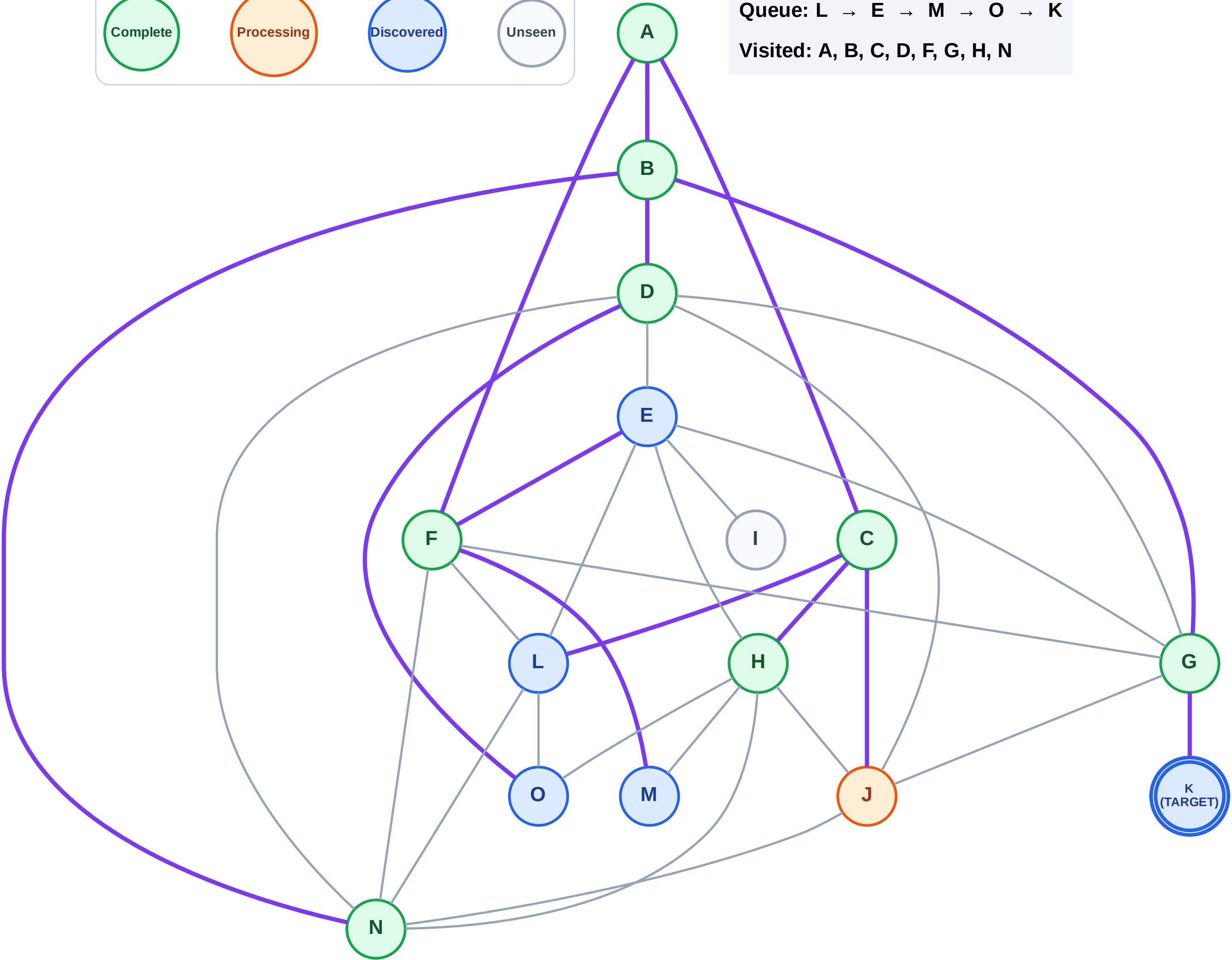
Processing

Discovered

Unseen

Queue: L → E → M → O → K

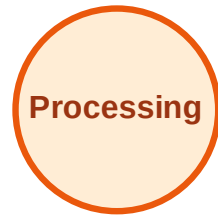
Visited: A, B, C, D, F, G, H, N



BFS Traversal

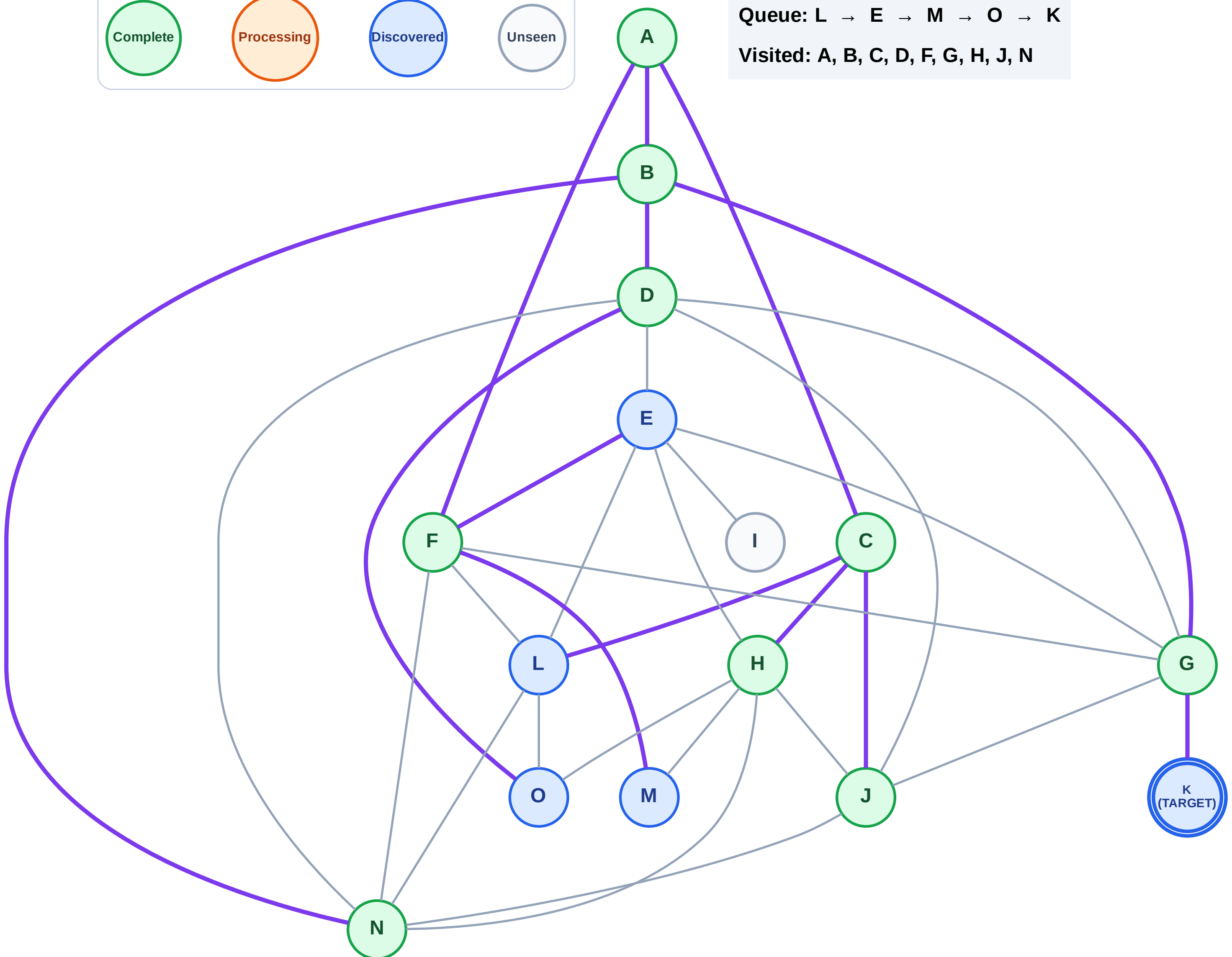
Step 19: No new neighbors from J

Legend



Queue: L → E → M → O → K

Visited: A, B, C, D, F, G, H, J, N



BFS Traversal

Step 20: Process node L

Legend

Complete

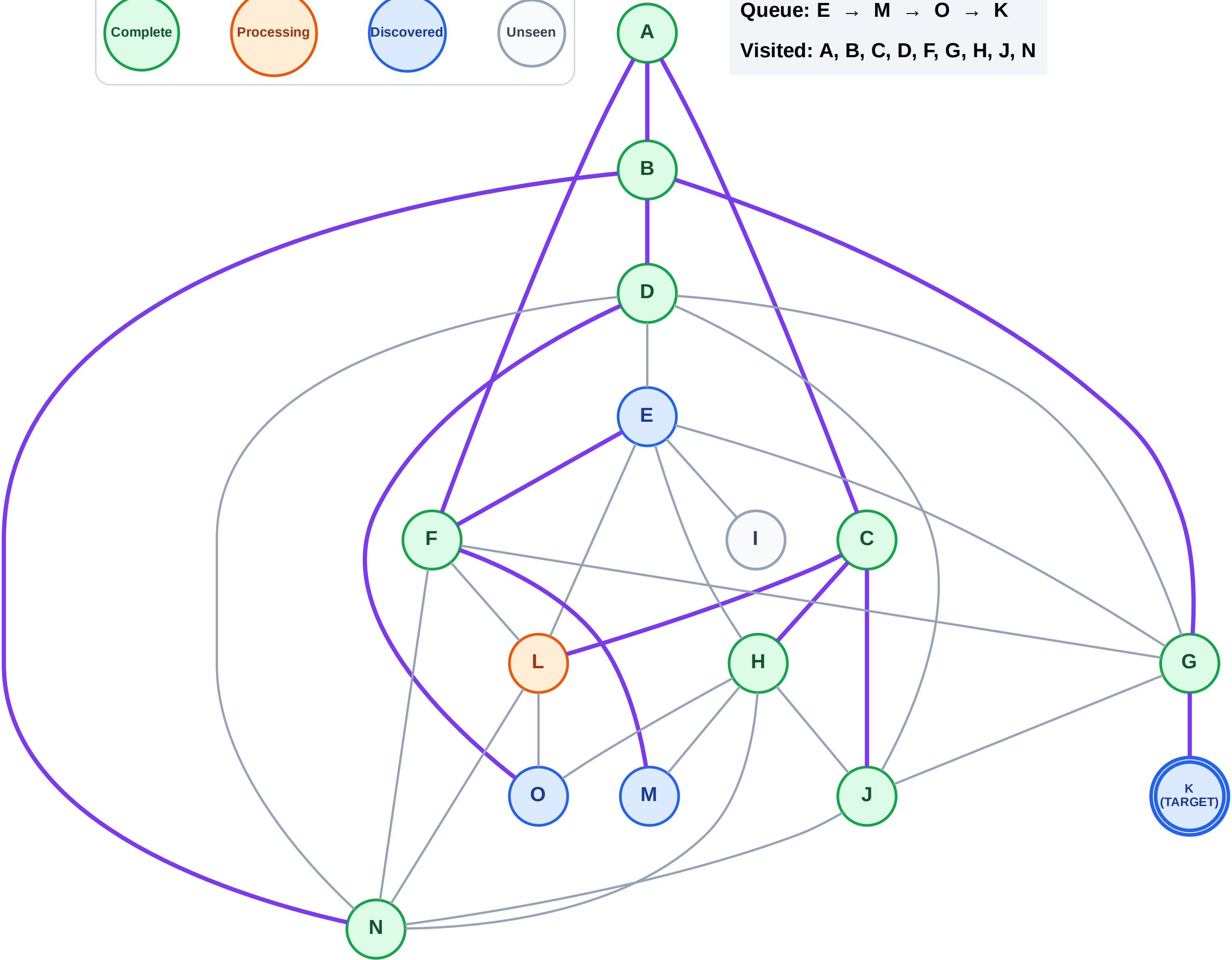
Processing

Discovered

Unseen

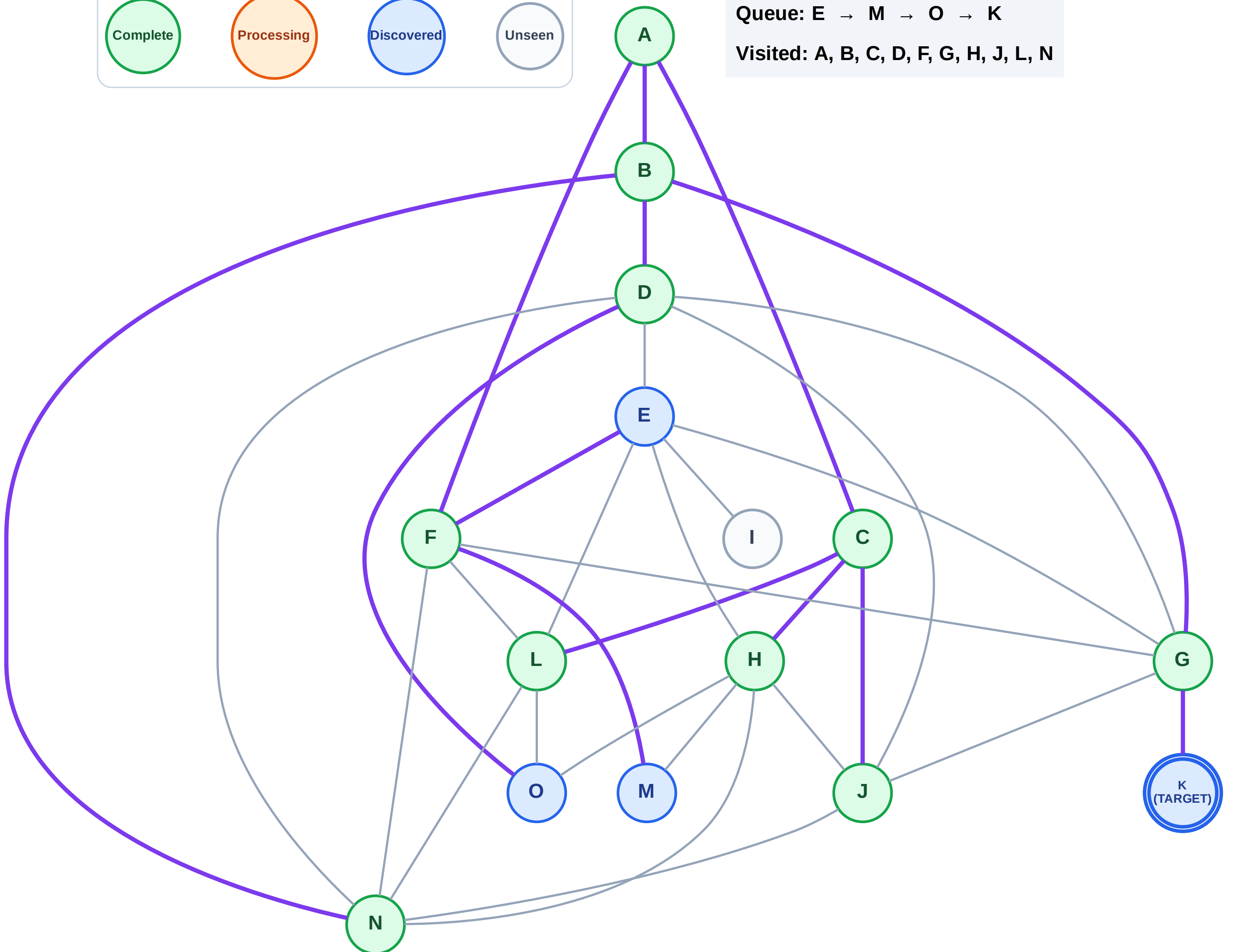
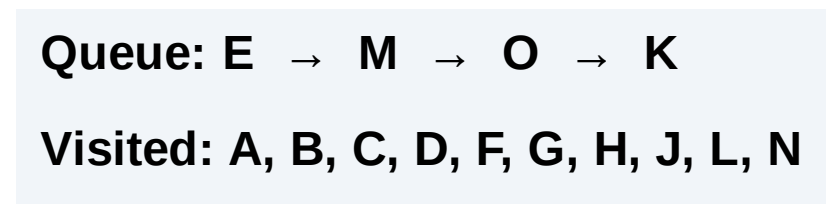
Queue: E → M → O → K

Visited: A, B, C, D, F, G, H, J, N



Step 21: No new neighbors from L

Step 21: No new neighbors from L

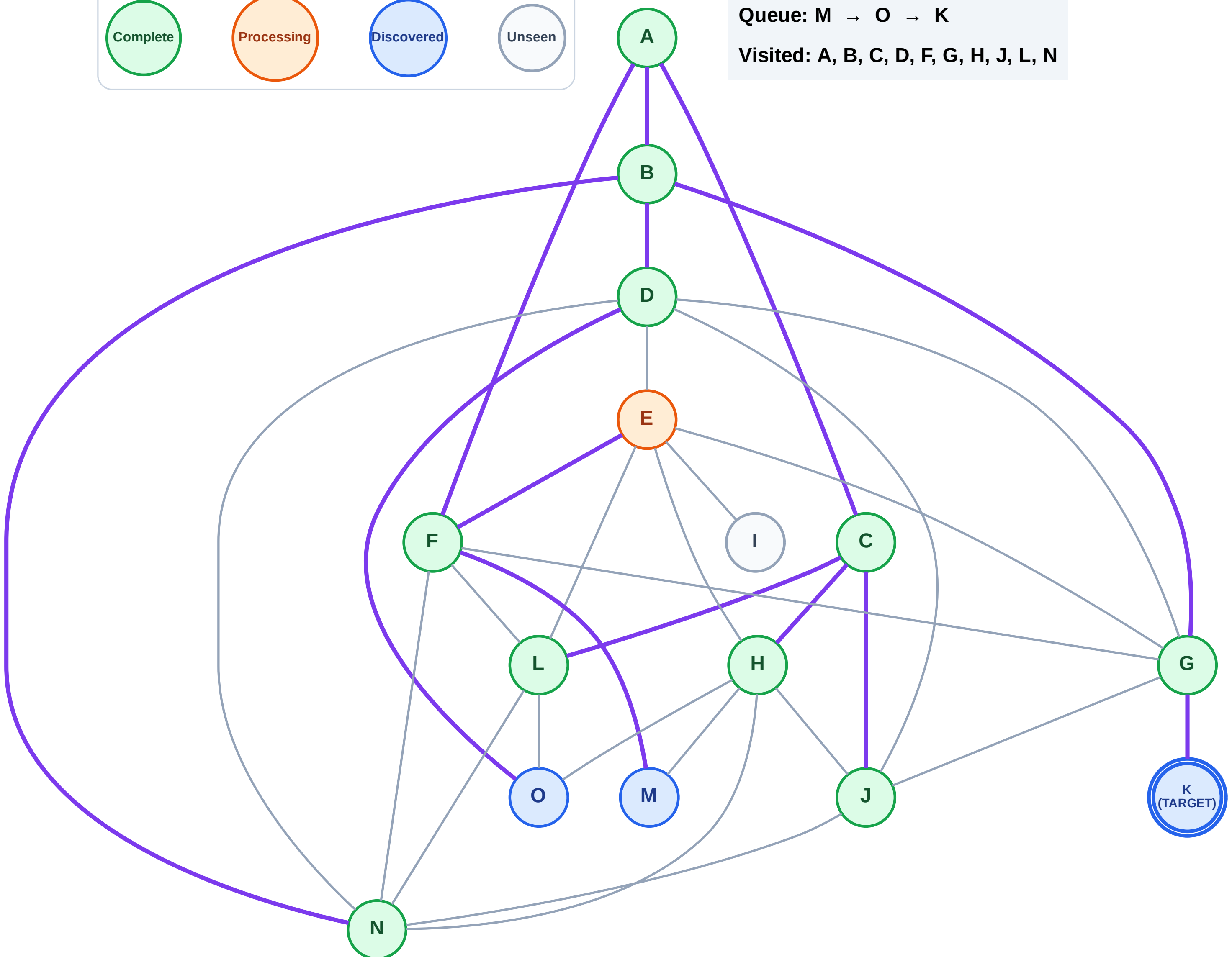


Step 22: Process node E

Legend

Unseen

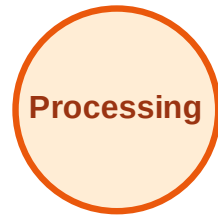
Visited: A, B, C, D, F, G, H, J, L, N



BFS Traversal

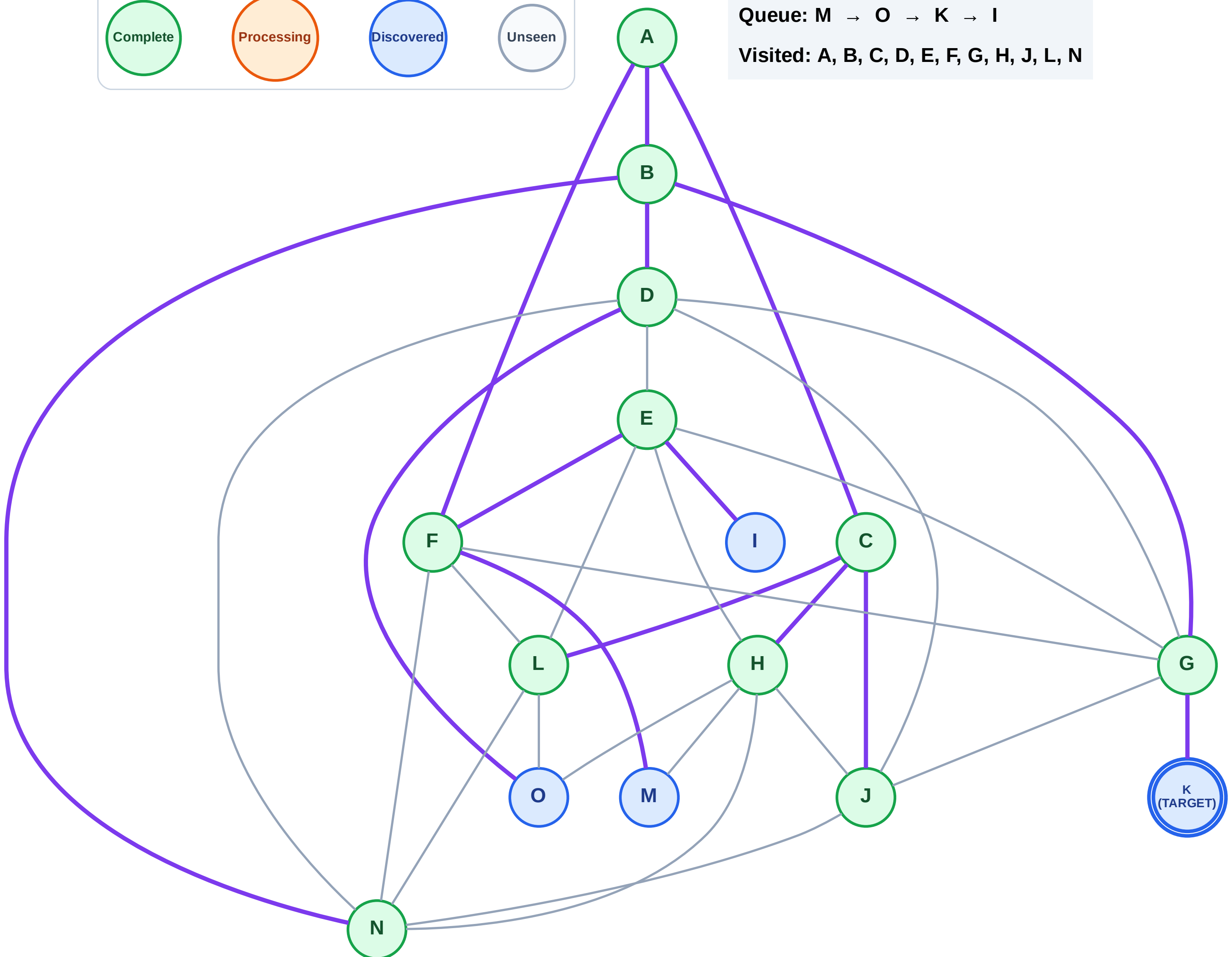
Step 23: Discover I from E

Legend

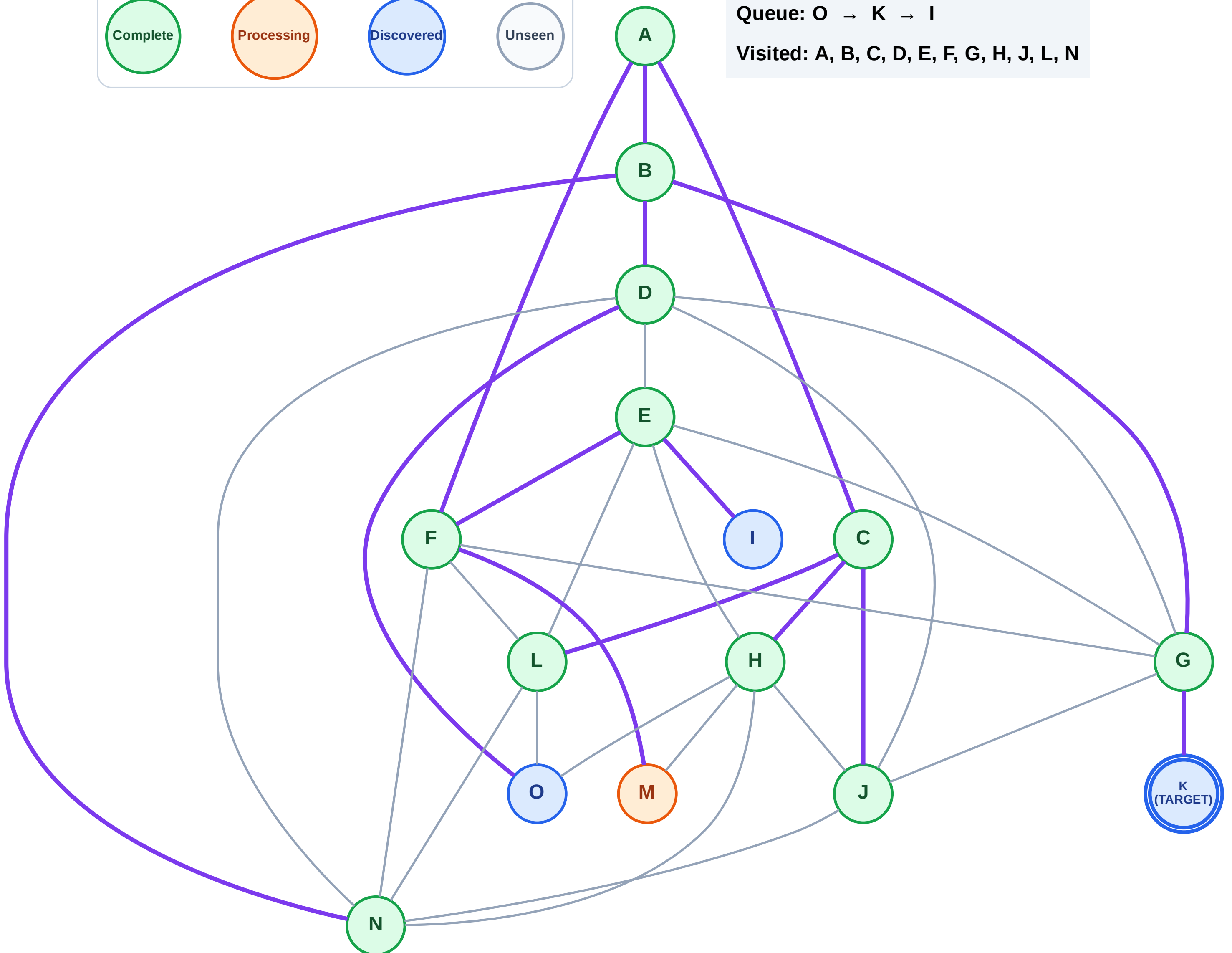


Queue: M → O → K → I

Visited: A, B, C, D, E, F, G, H, J, L, N



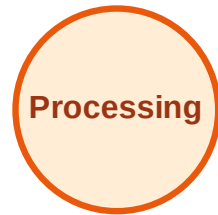
Step 24: Process node M



BFS Traversal

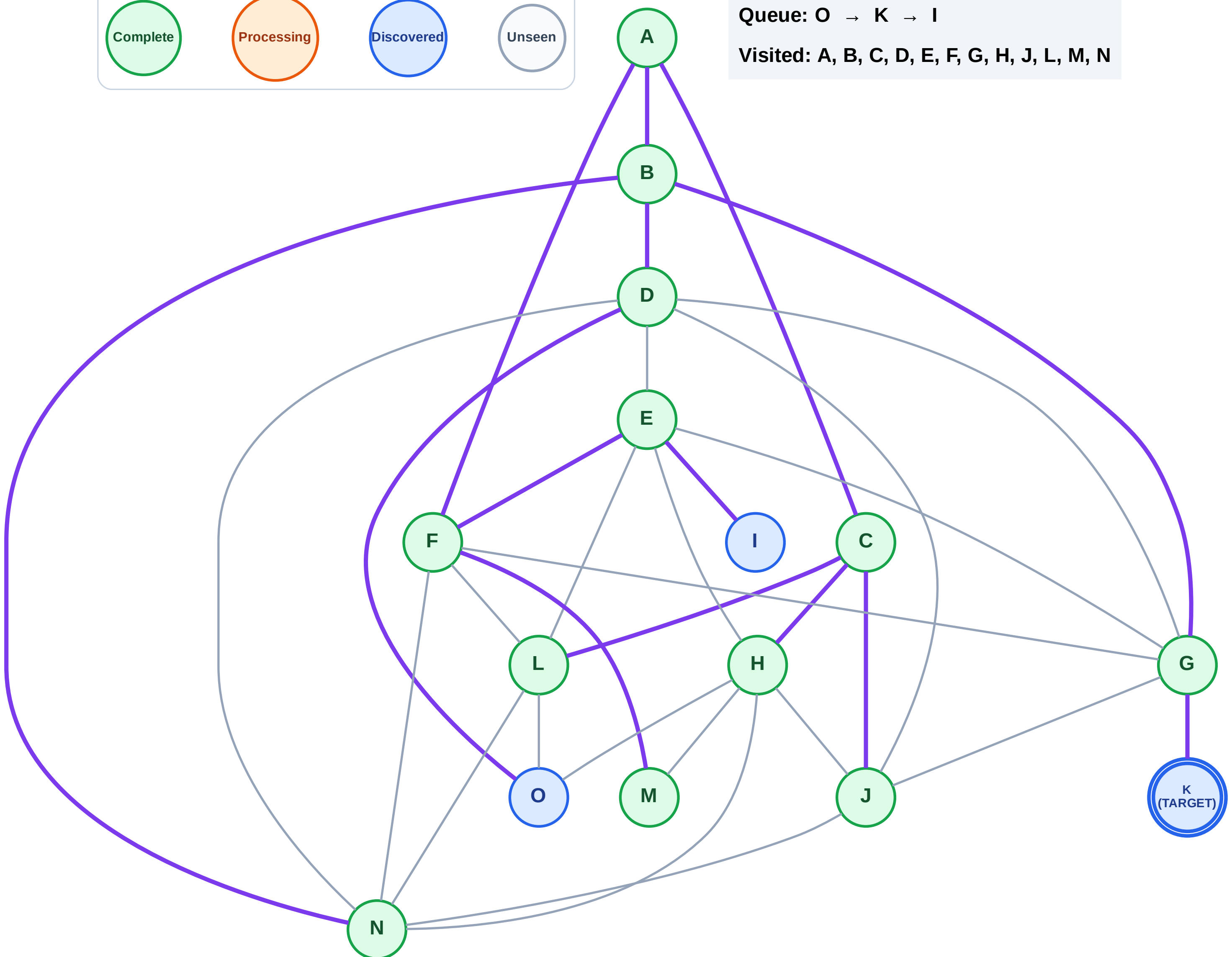
Step 25: No new neighbors from M

Legend



Queue: O → K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, N



BFS Traversal

Step 26: Process node O

Legend

Complete

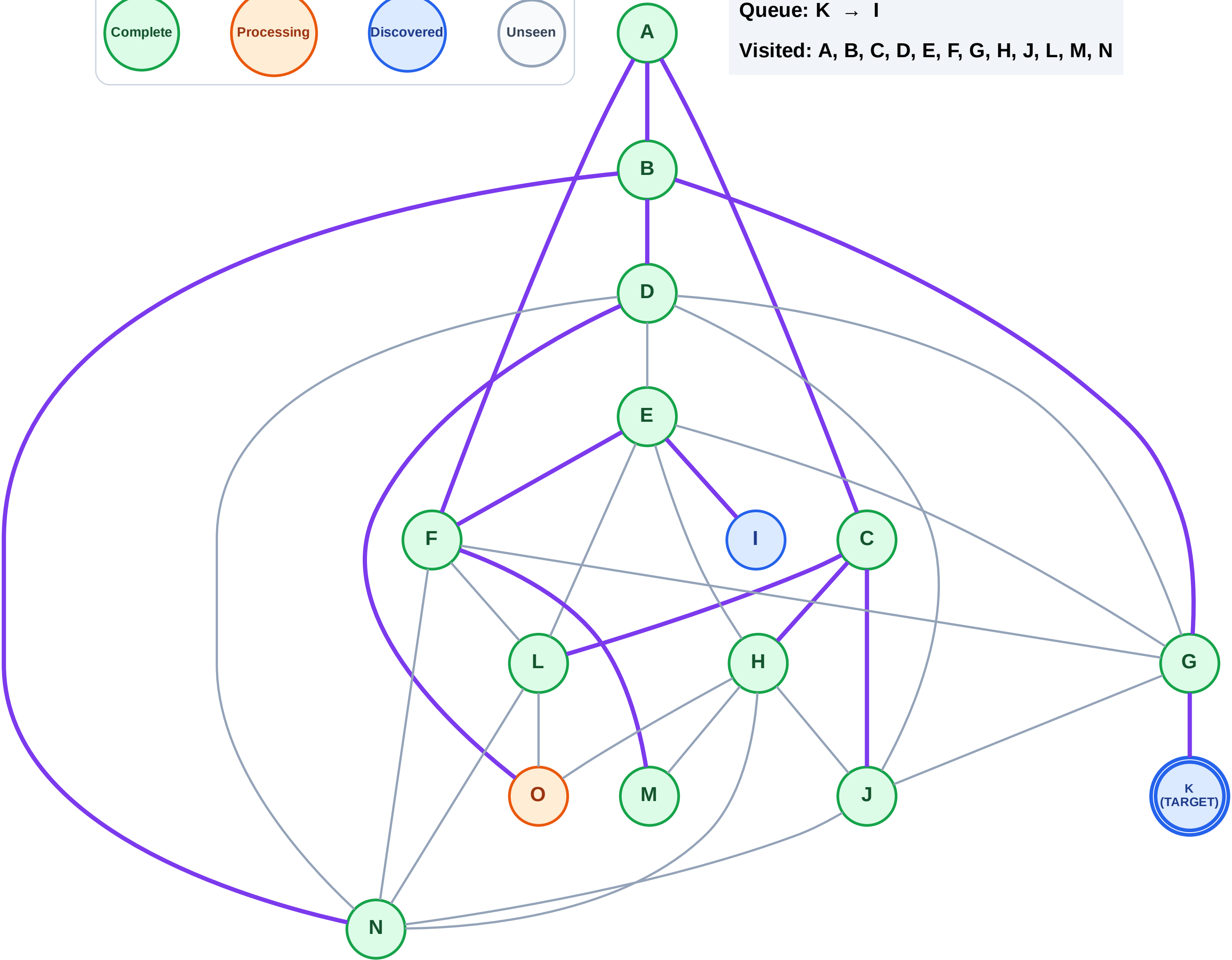
Processing

Discovered

Unseen

Queue: K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, N

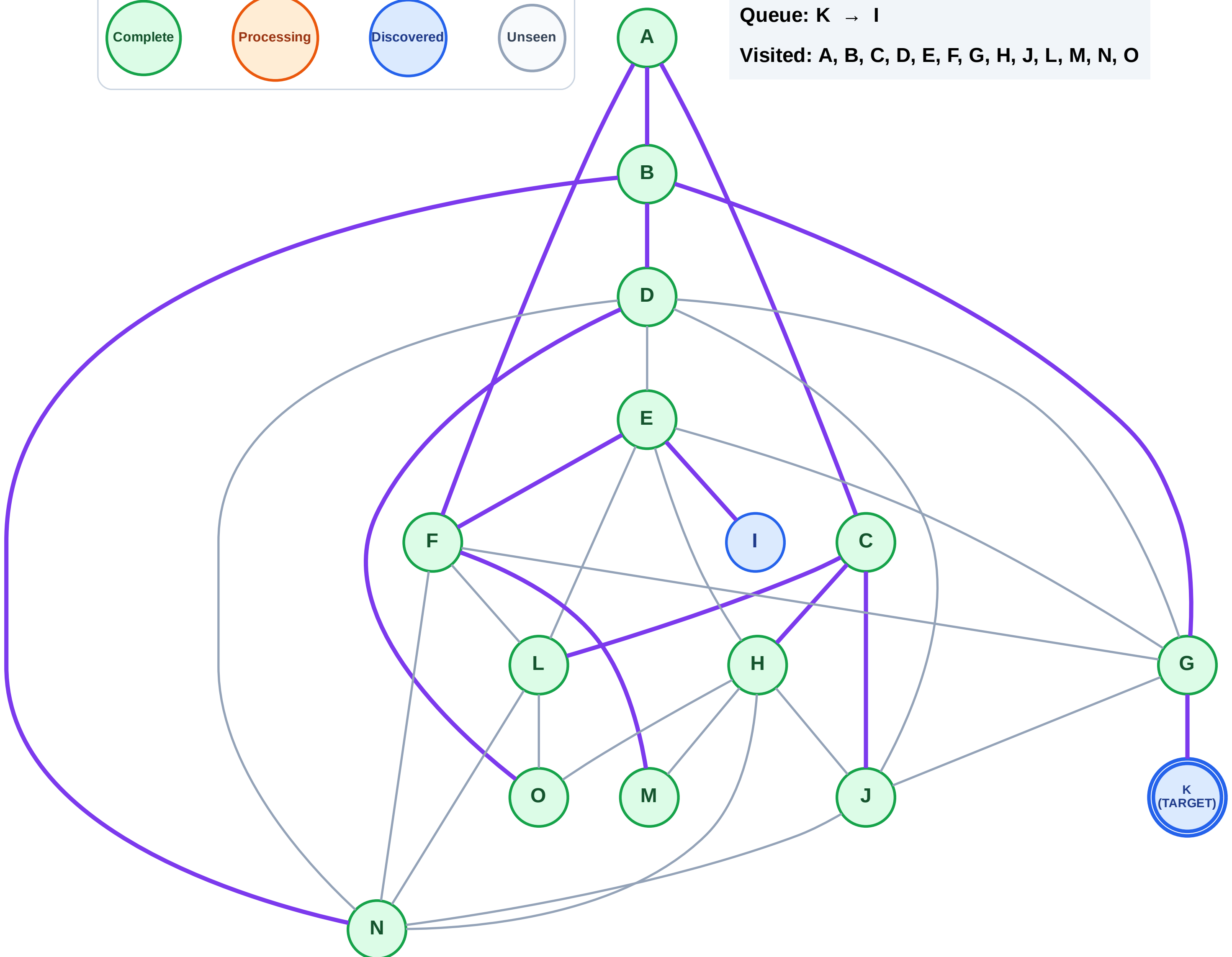


Step 27: No new neighbors from O

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

Step 28: Process node K

Legend

Complete

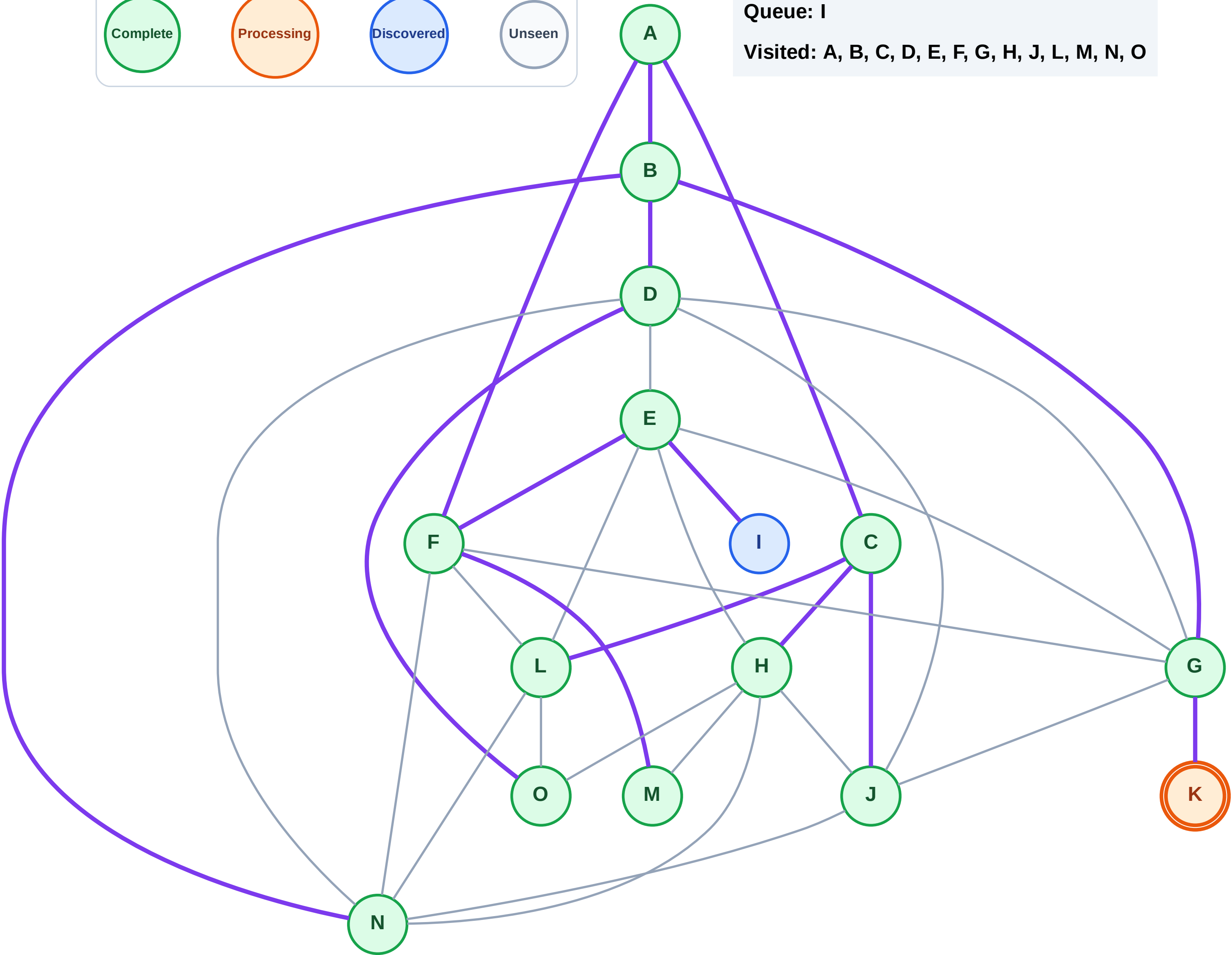
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → B → G → K

Legend

Complete

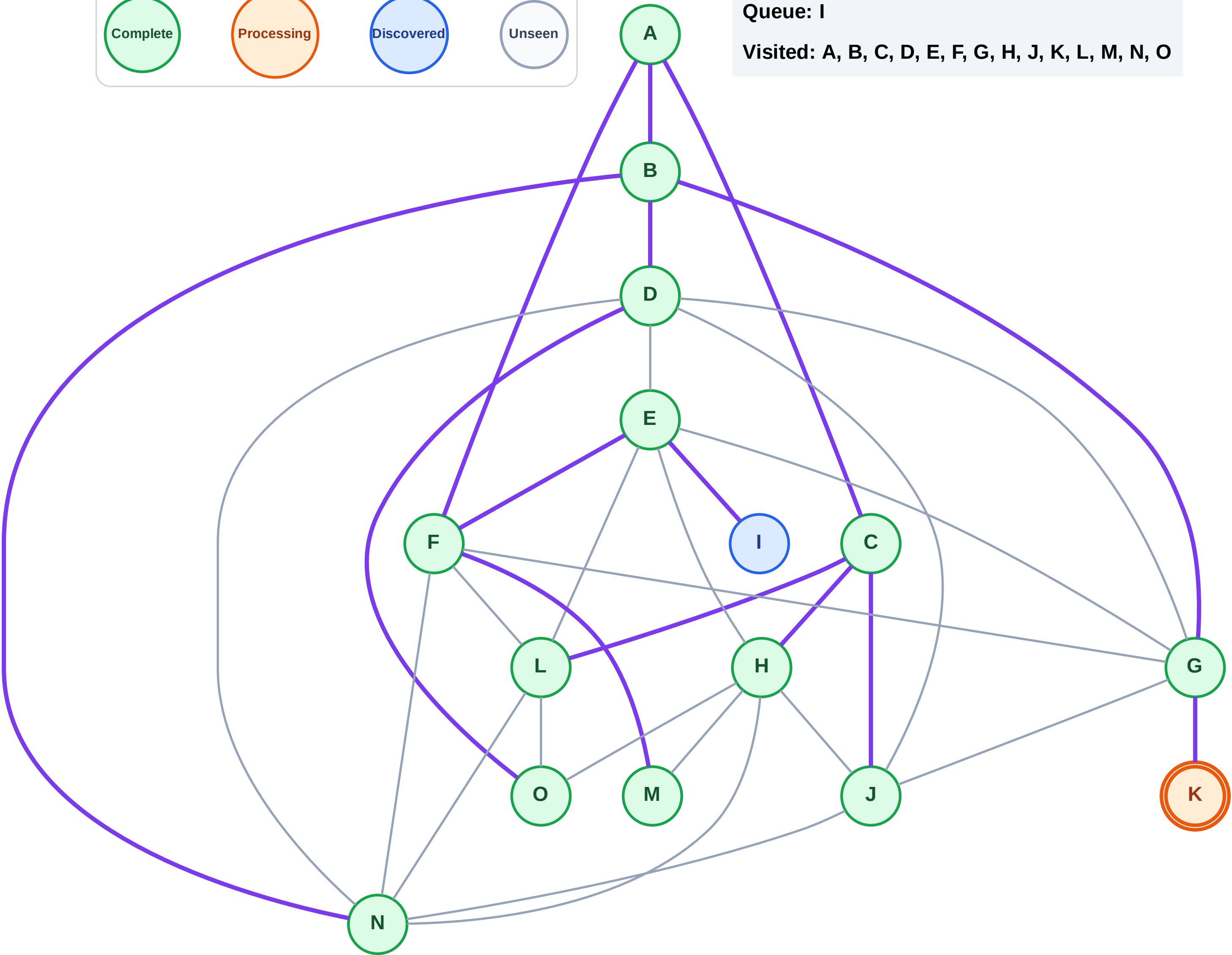
Processing

Discovered

Unseen

Queue: I

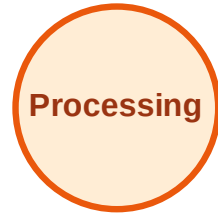
Visited: A, B, C, D, E, F, G, H, J, K, L, M, N, O



DFS Traversal

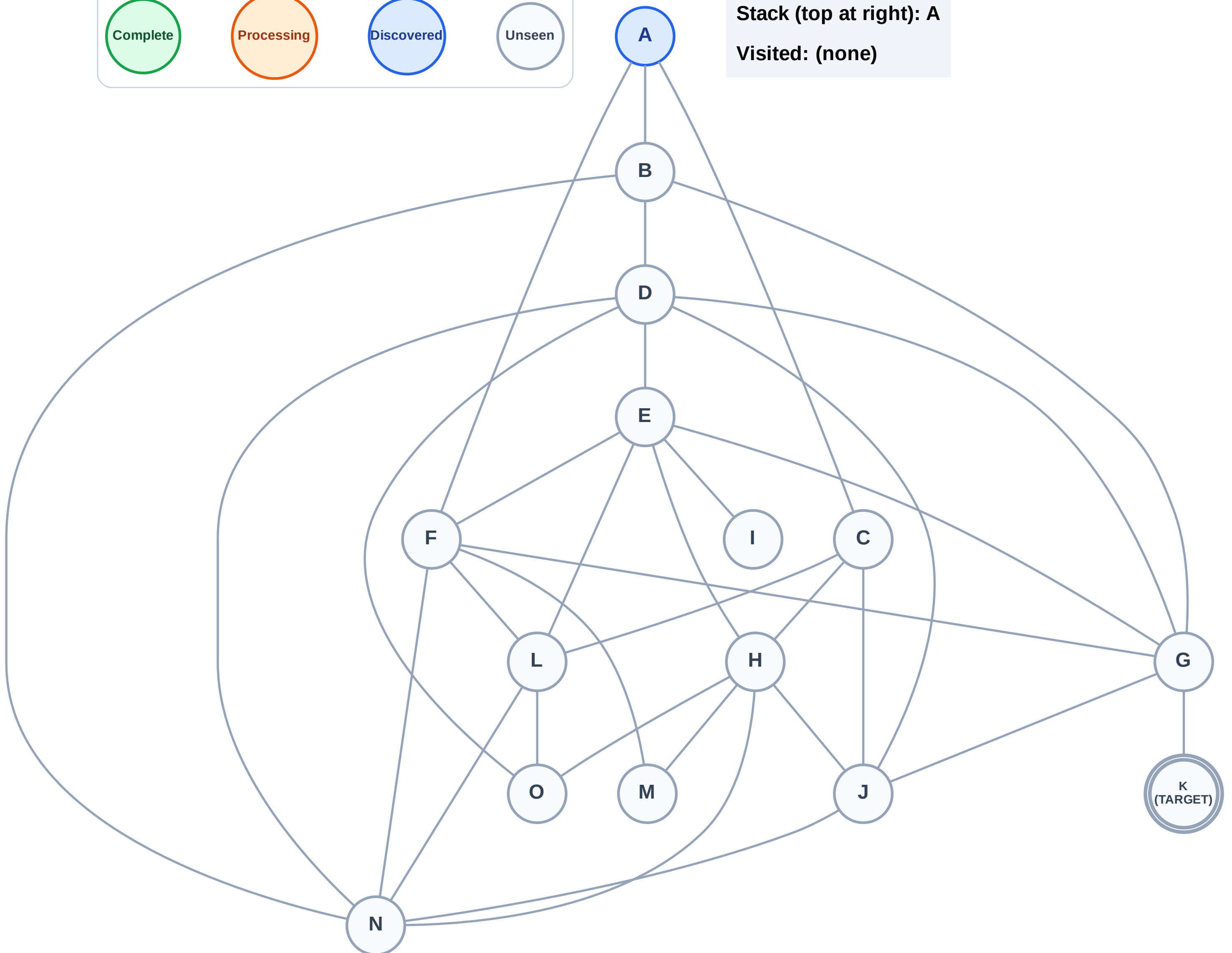
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

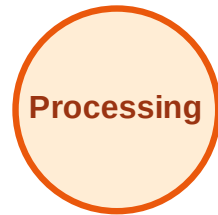
Visited: (none)



DFS Traversal

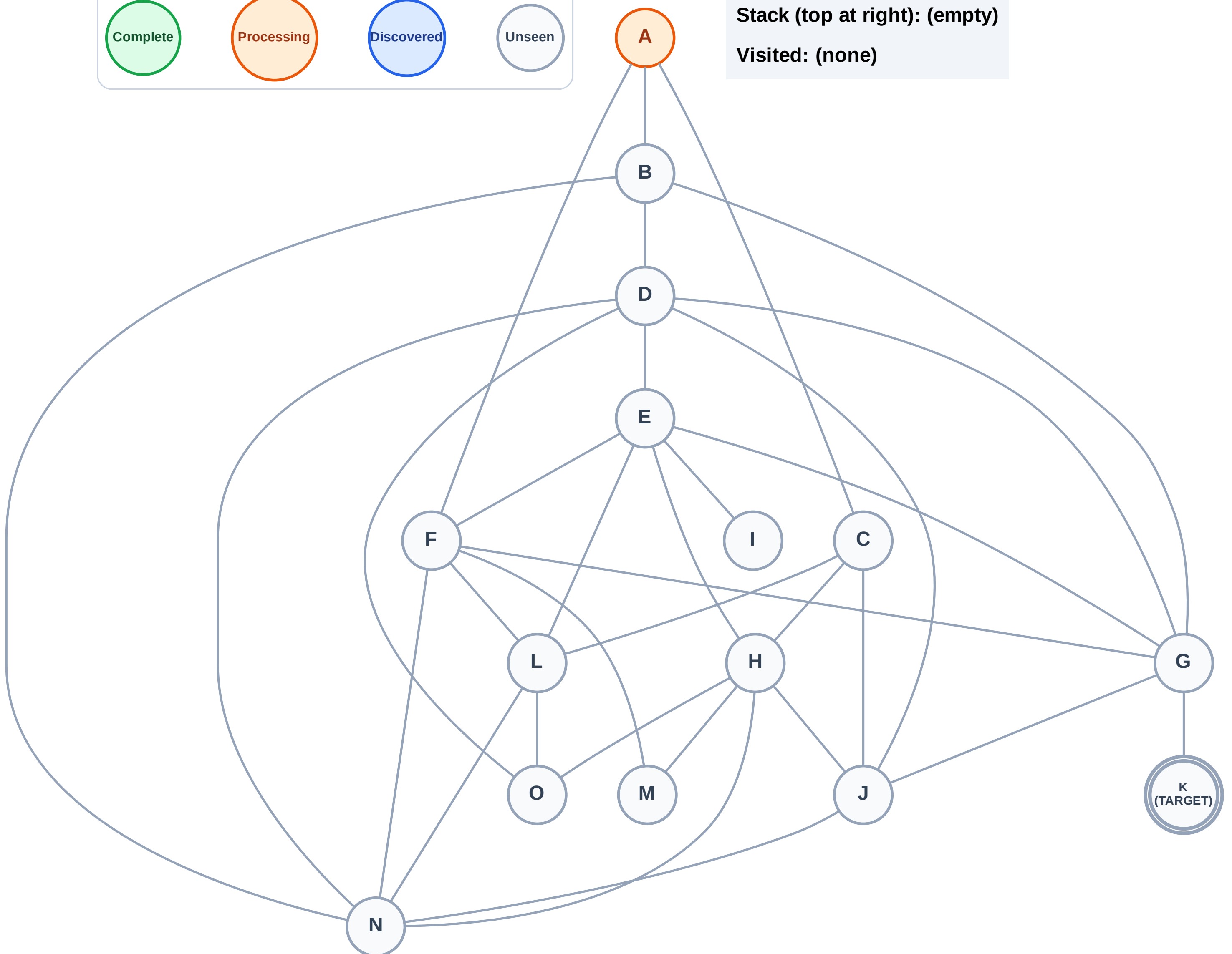
Step 2: Process node A

Legend



Stack (top at right): (empty)

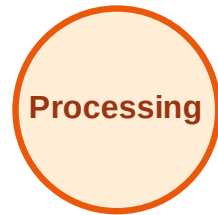
Visited: (none)



DFS Traversal

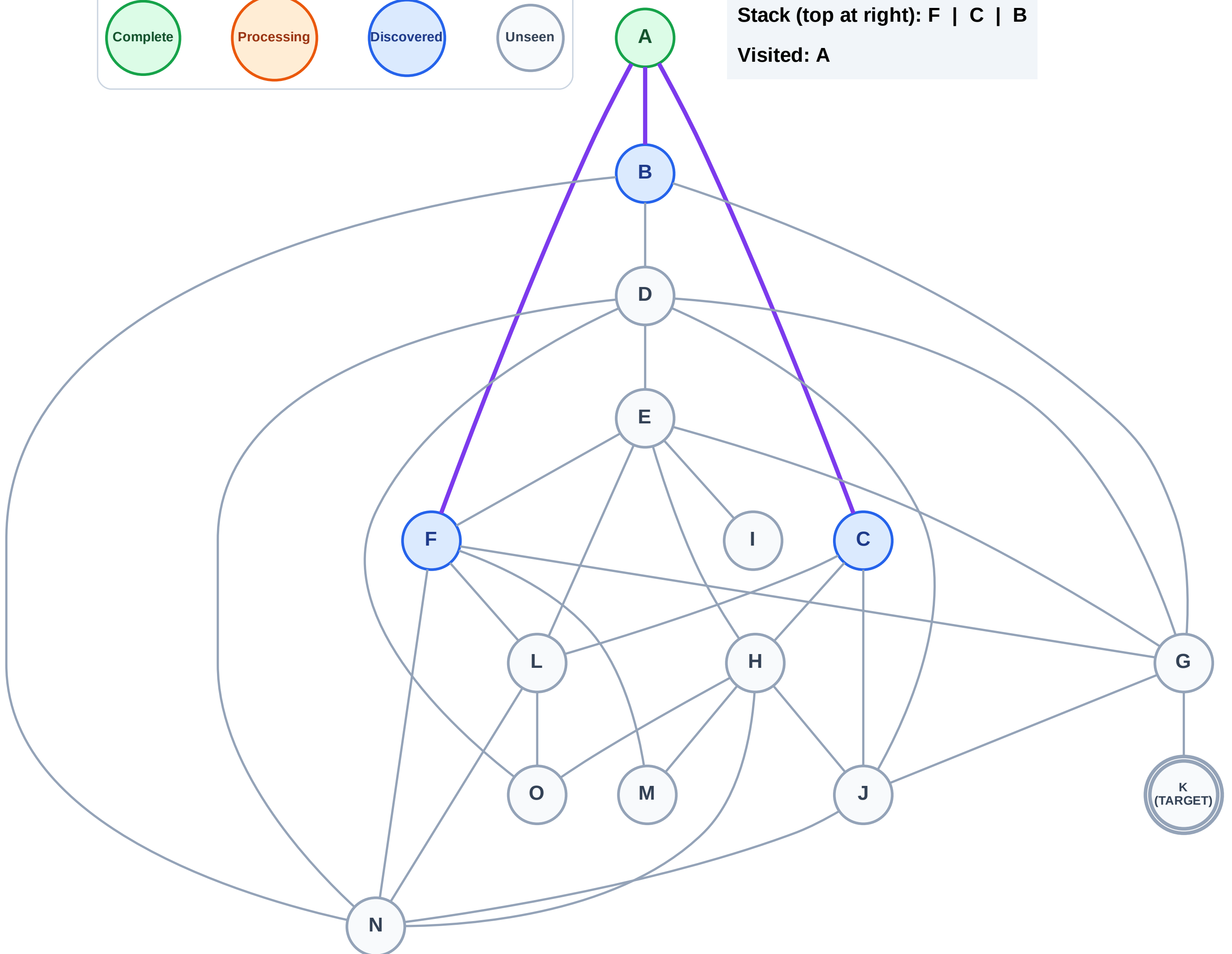
Step 3: Discover F, C, B from A

Legend



Stack (top at right): F | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

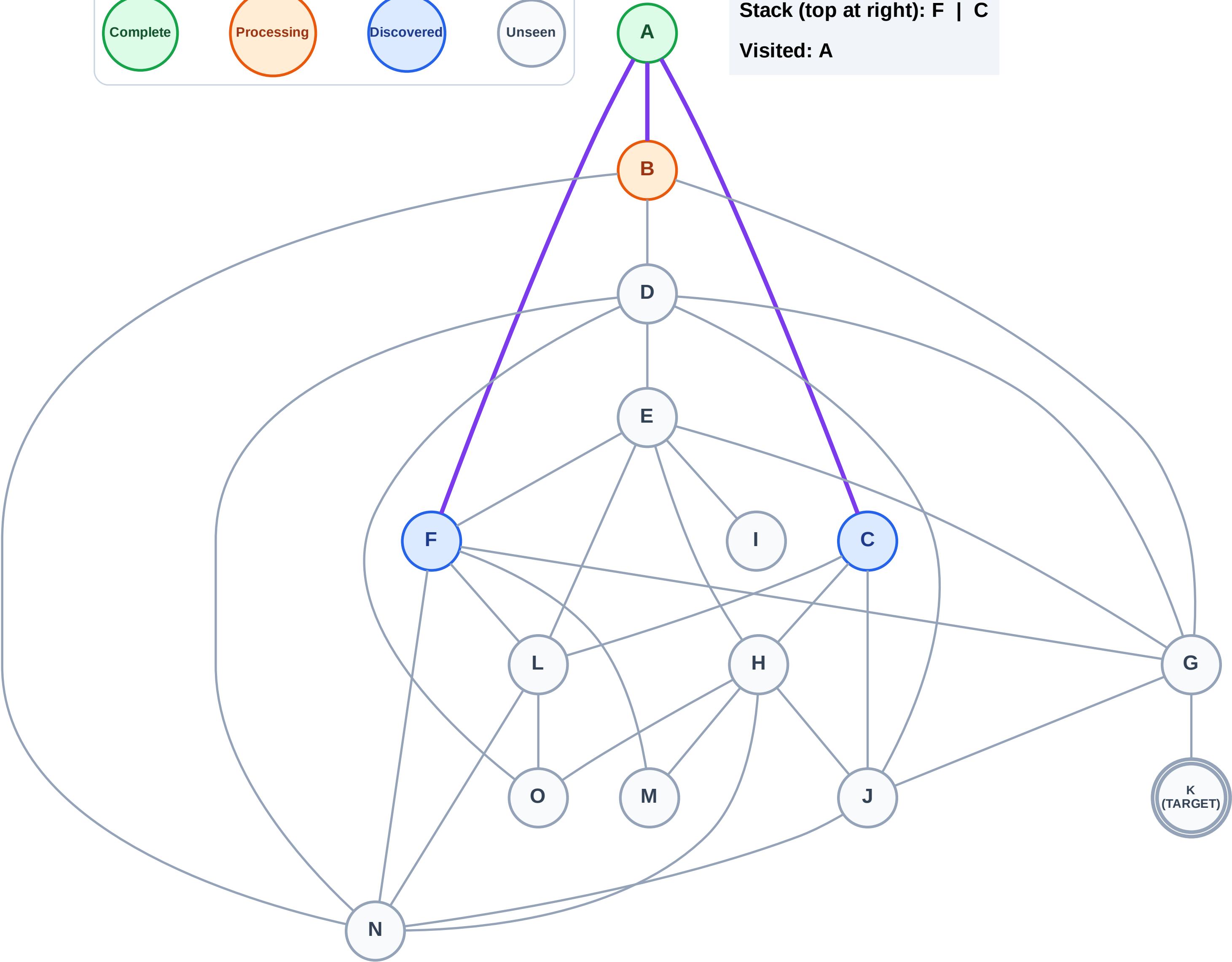
Complete

Processing

Discovered

Unseen

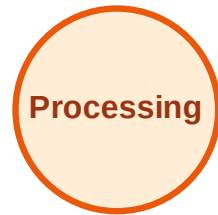
Stack (top at right): F | C
Visited: A



DFS Traversal

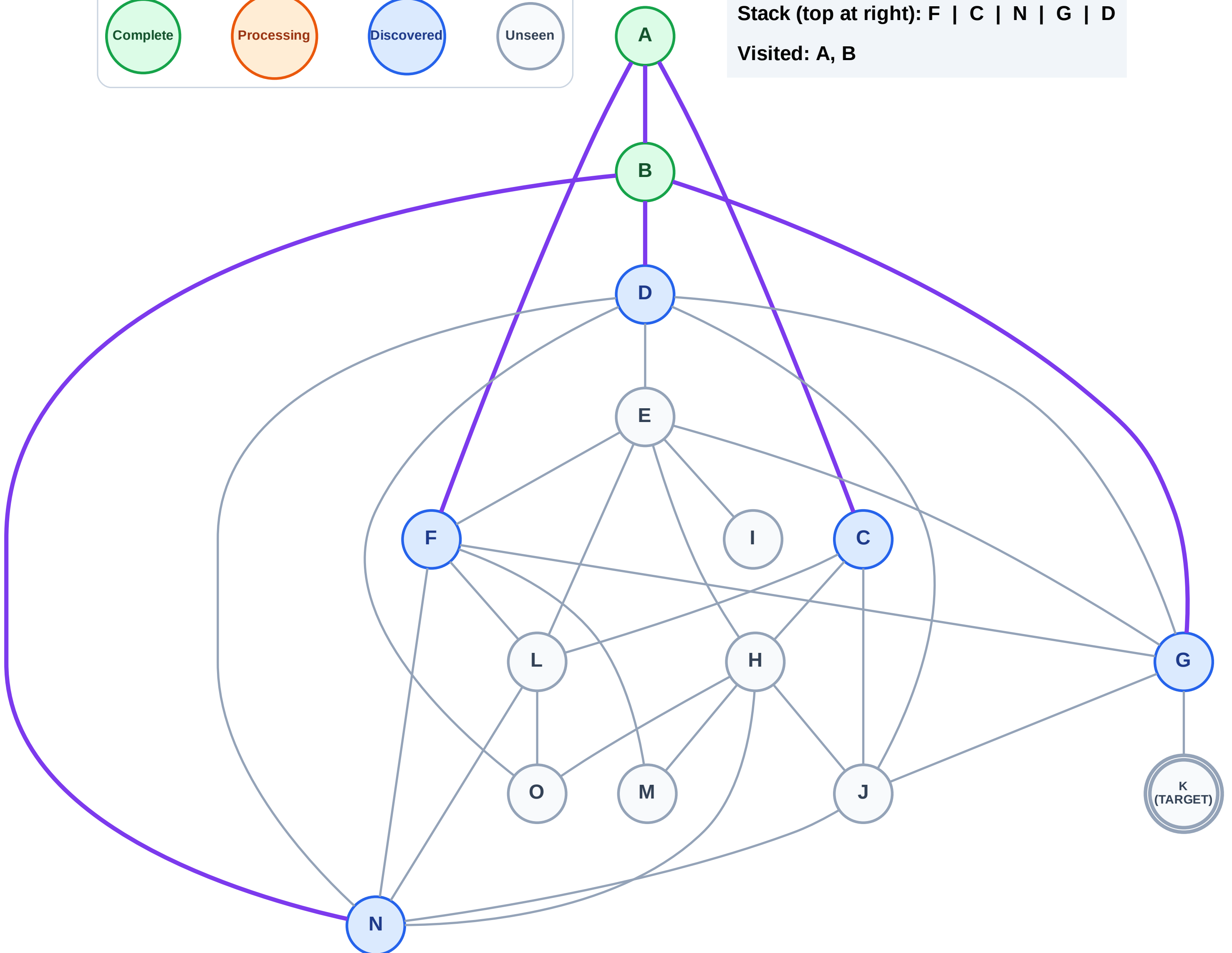
Step 5: Discover N, G, D from B

Legend



Stack (top at right): F | C | N | G | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

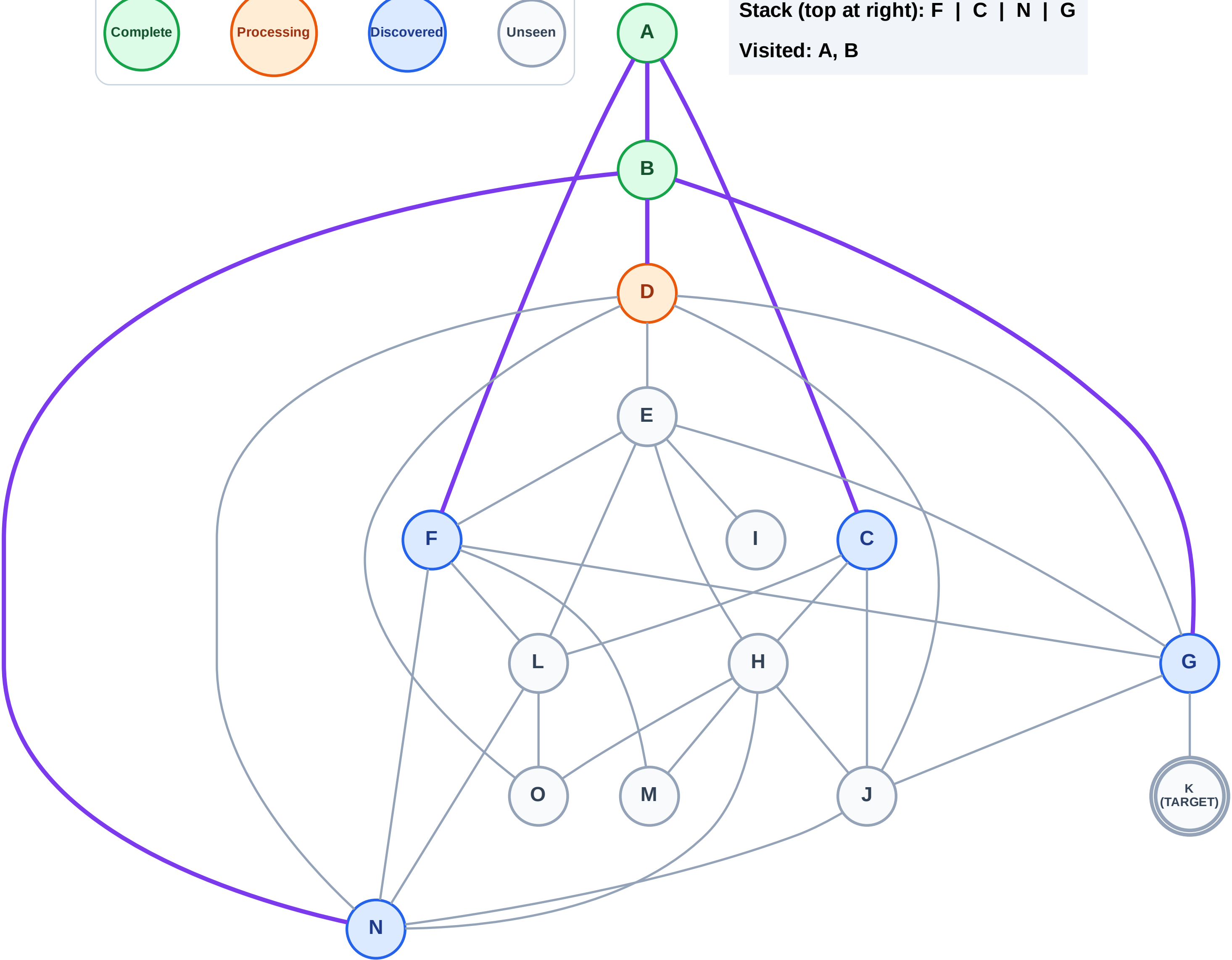
Complete

Processing

Discovered

Unseen

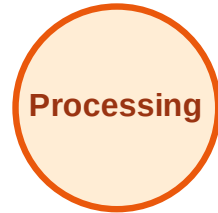
Stack (top at right): F | C | N | G
Visited: A, B



DFS Traversal

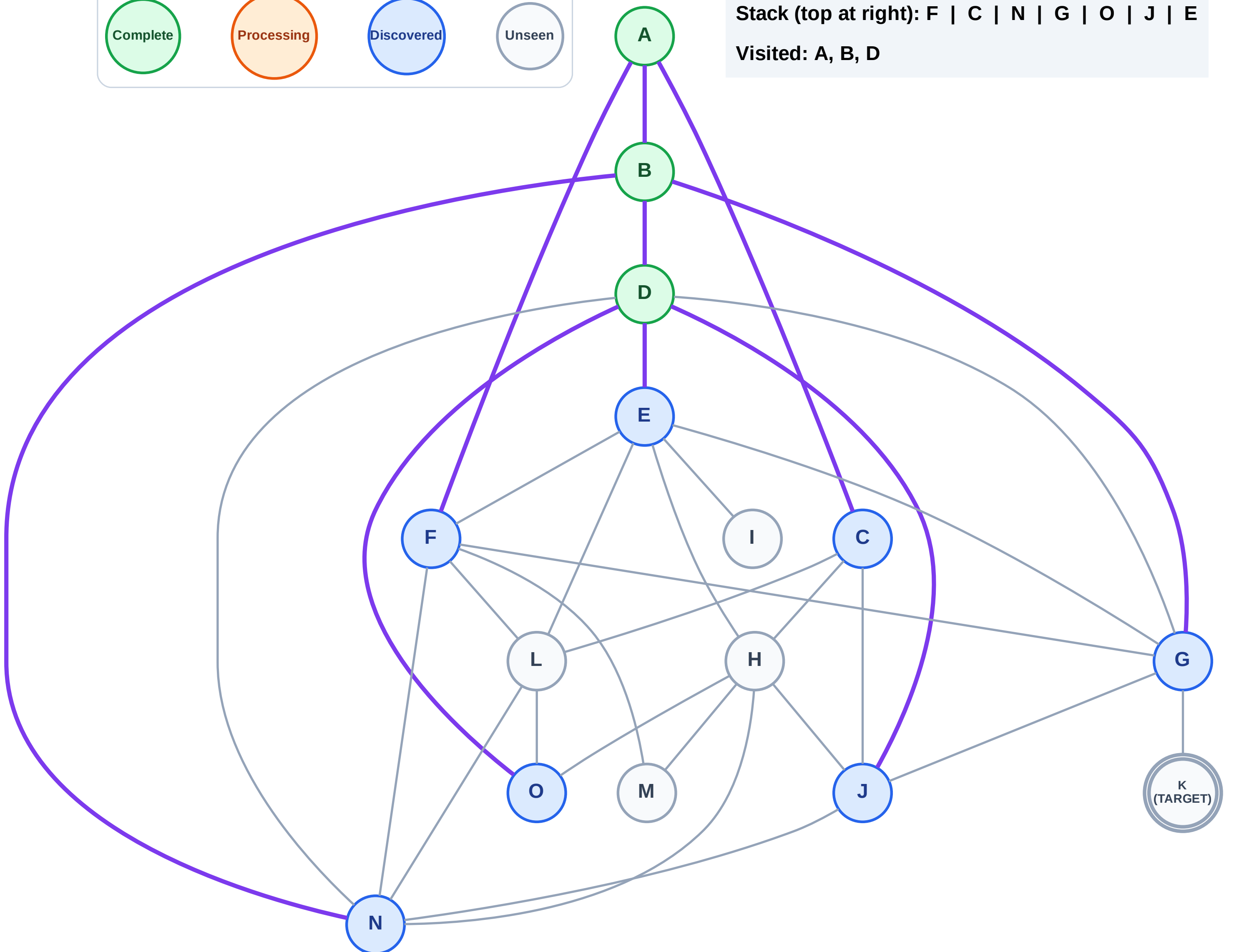
Step 7: Discover O, J, E from D

Legend



Stack (top at right): F | C | N | G | O | J | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

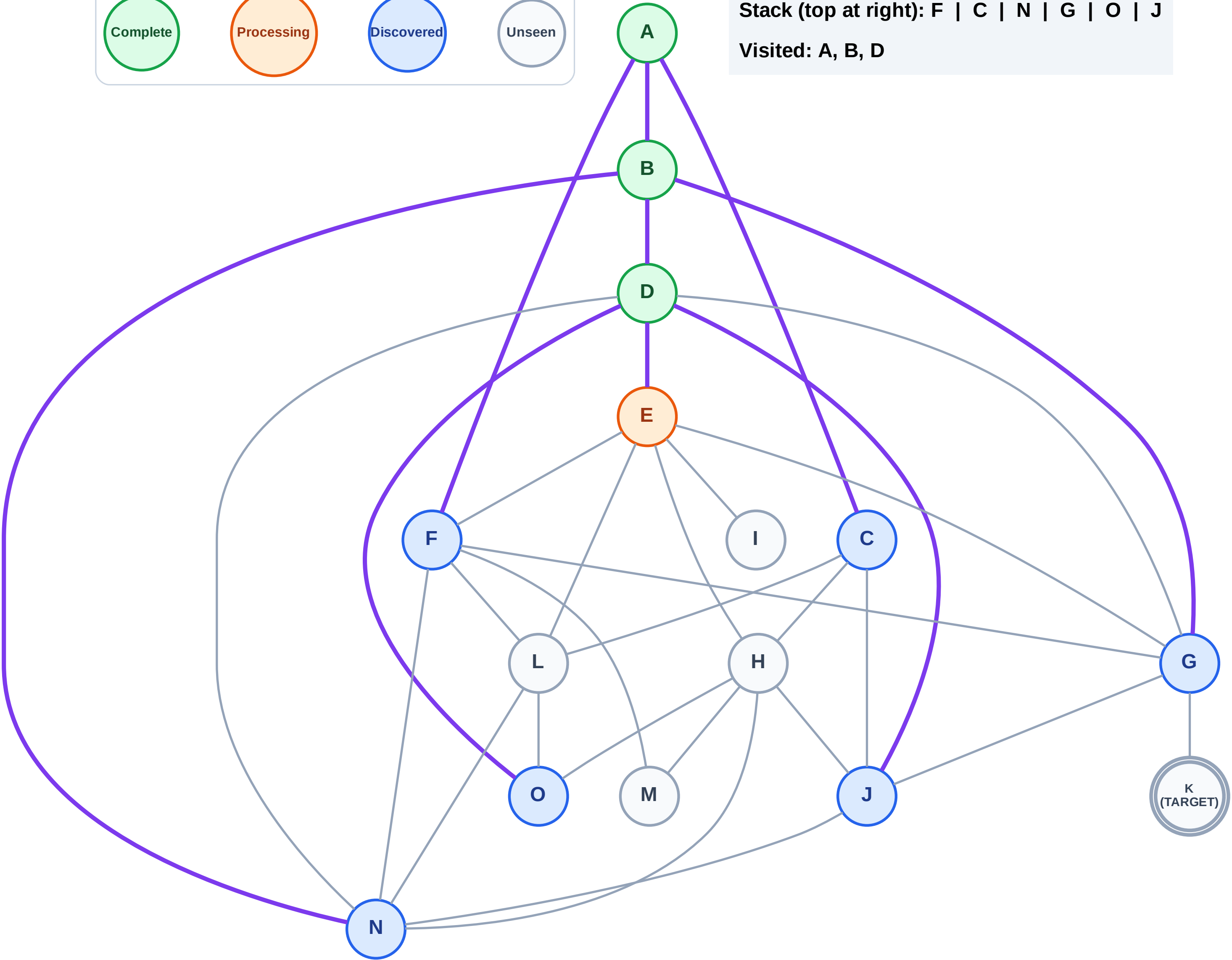
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J

Visited: A, B, D



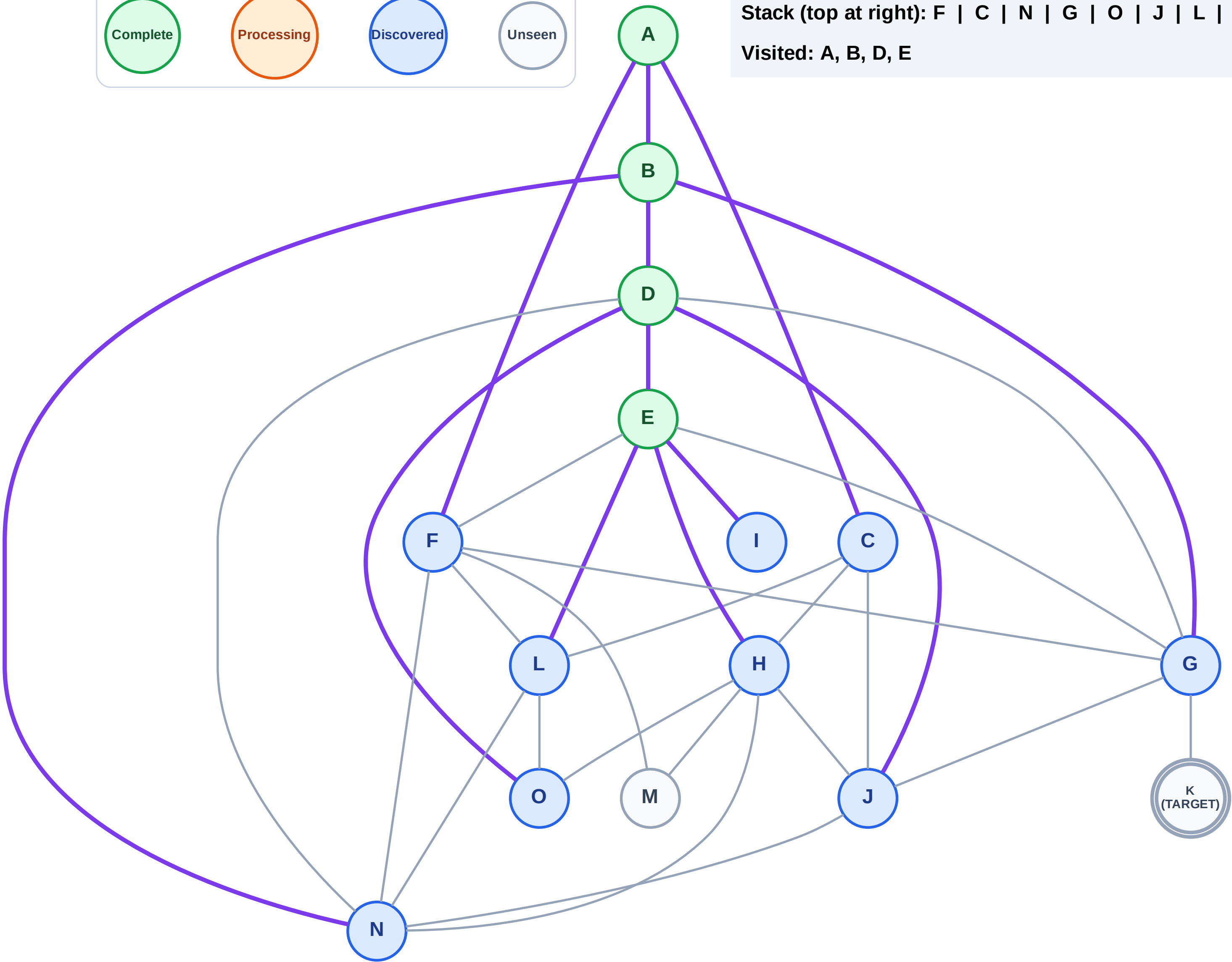
Step 9: Discover L, I, H from E

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Visited: A, B, D, E



DFS Traversal

Step 10: Process node H

Legend

Complete

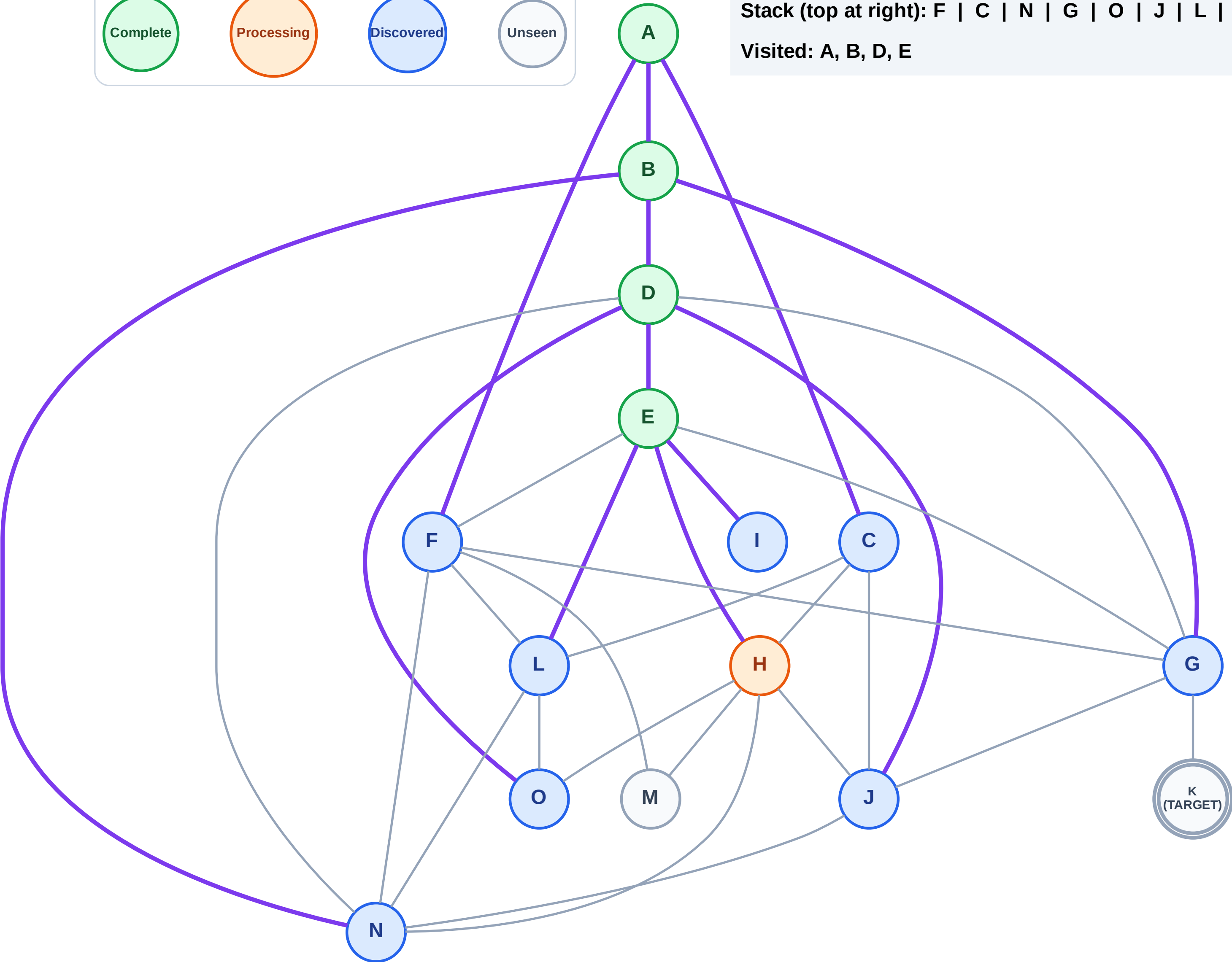
Processing

Discovered

Unseen

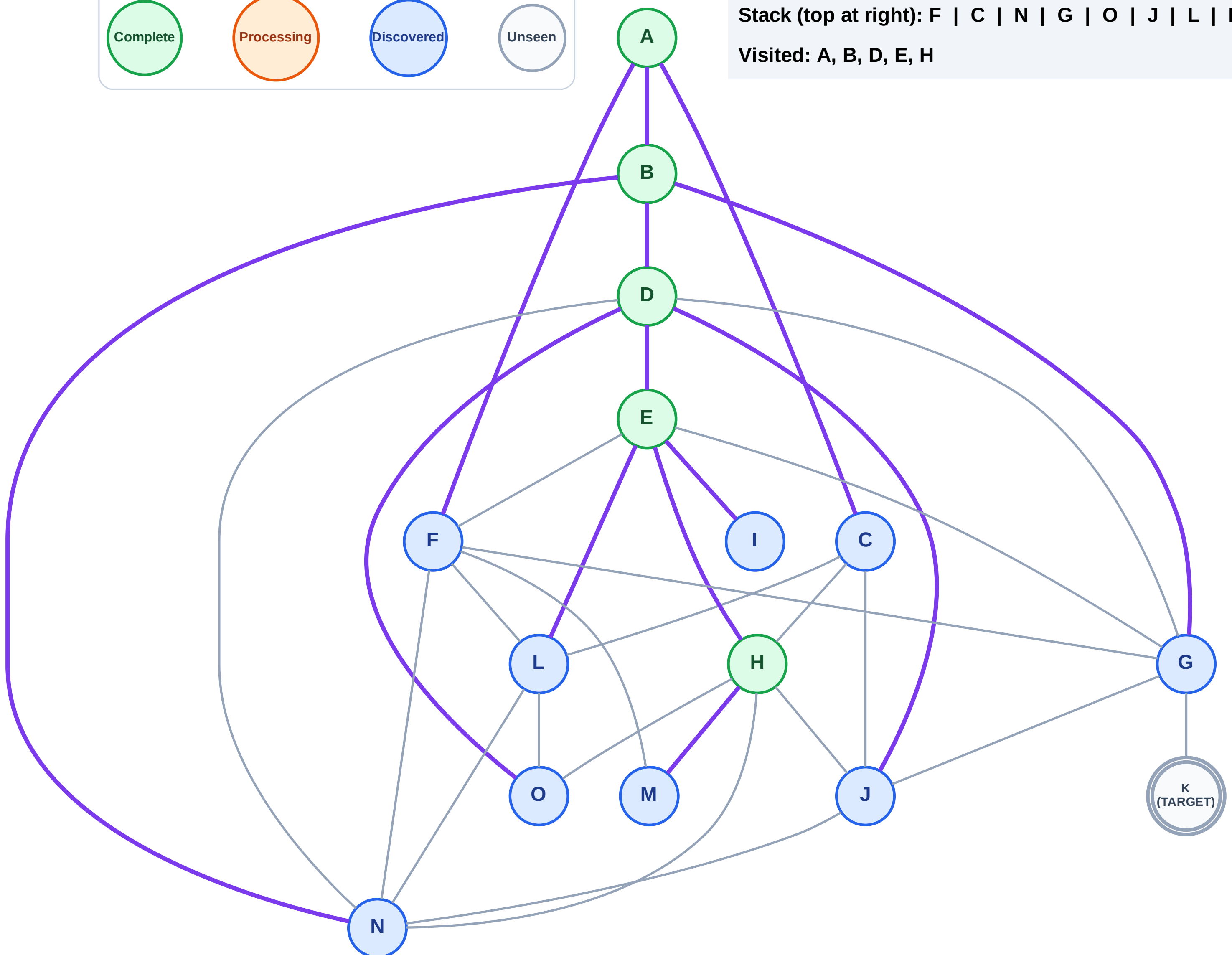
Stack (top at right): F | C | N | G | O | J | L | I

Visited: A, B, D, E

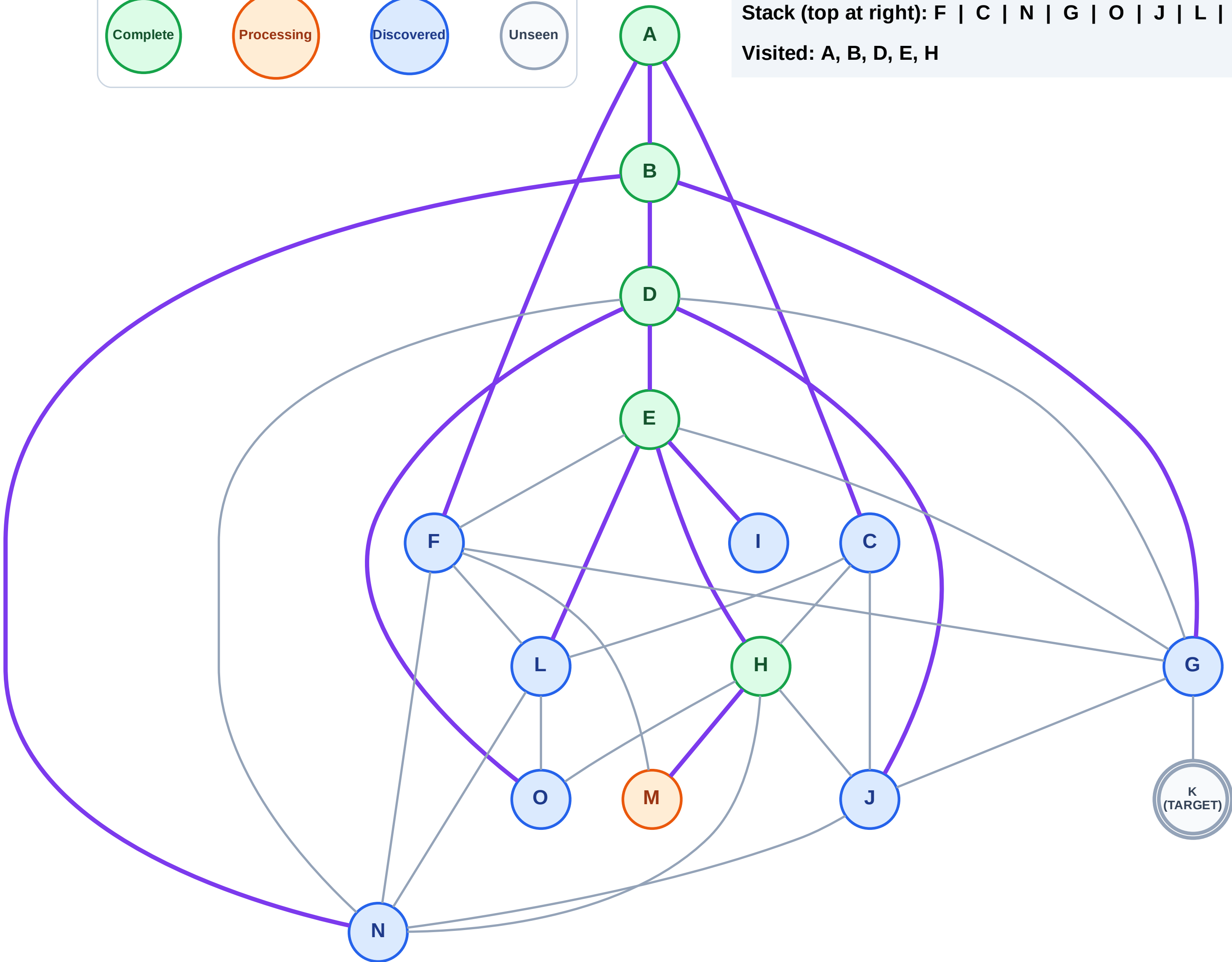


Step 11: Discover M from H

Step 11: Discover M from H



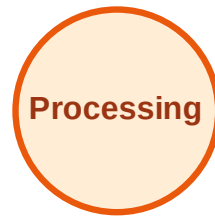
Step 12: Process node M



DFS Traversal

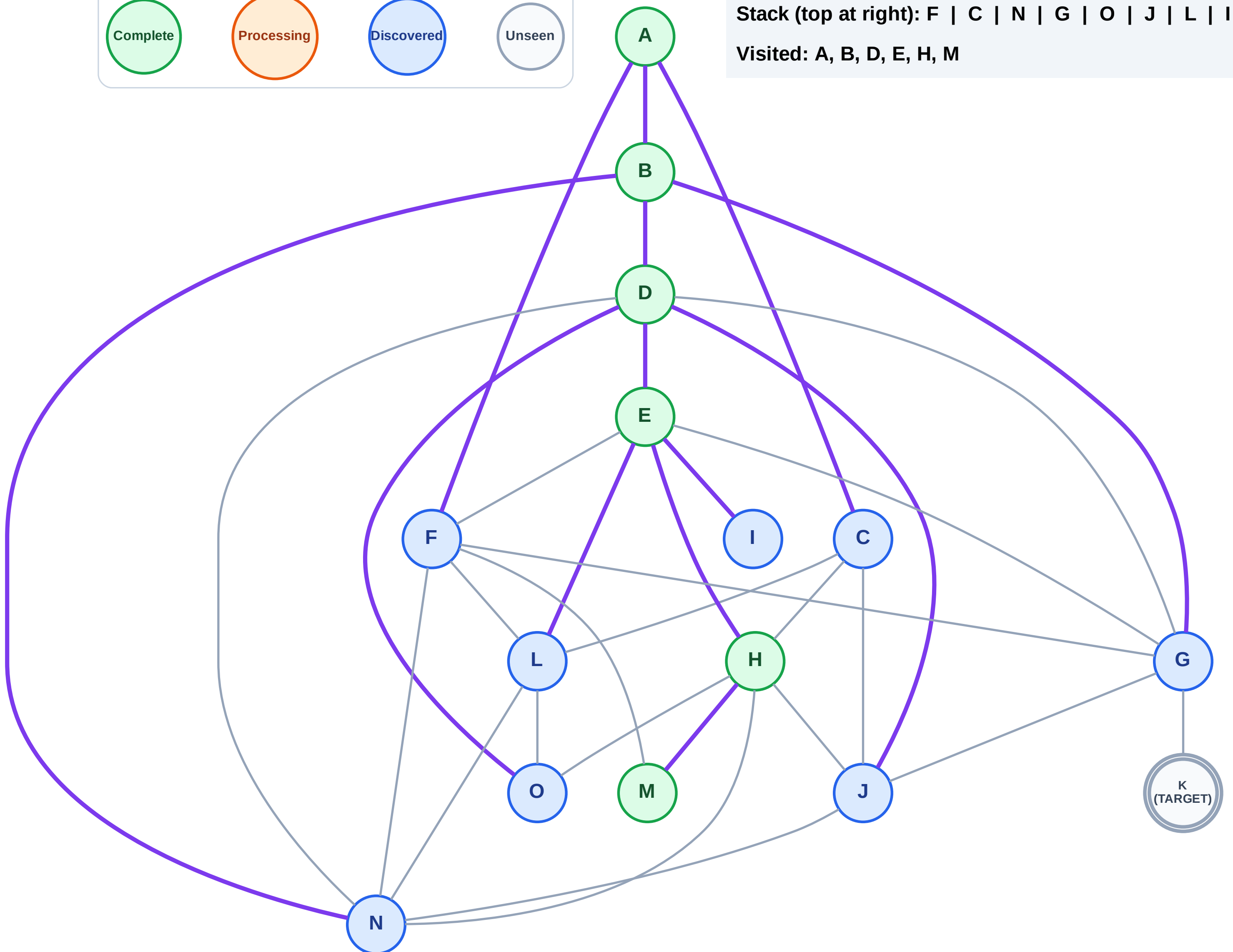
Step 13: No new neighbors from M

Legend



Stack (top at right): F | C | N | G | O | J | L | I

Visited: A, B, D, E, H, M



DFS Traversal

Step 14: Process node I

Legend

Complete

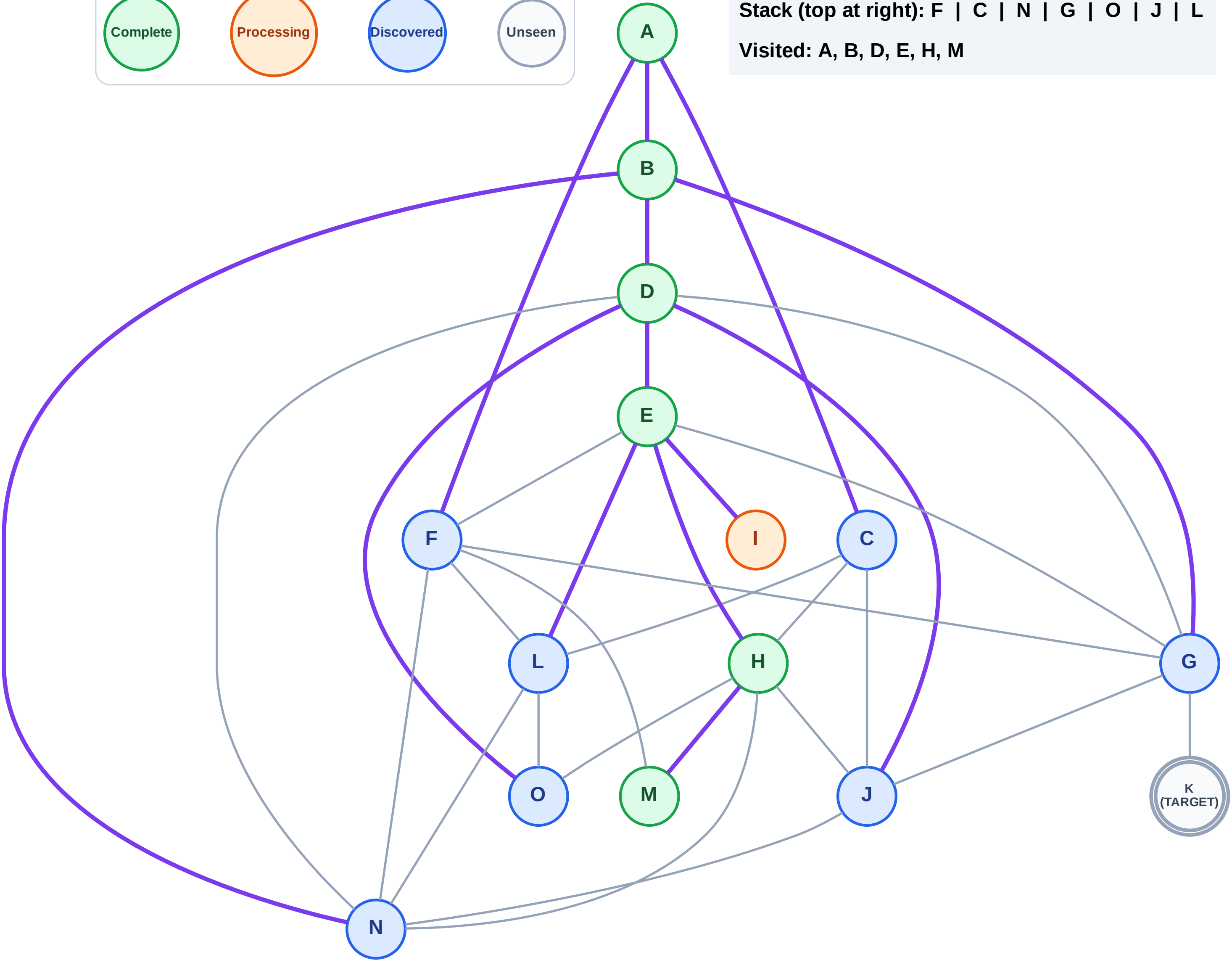
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J | L

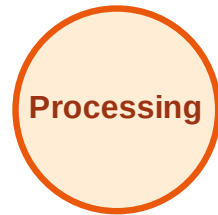
Visited: A, B, D, E, H, M



DFS Traversal

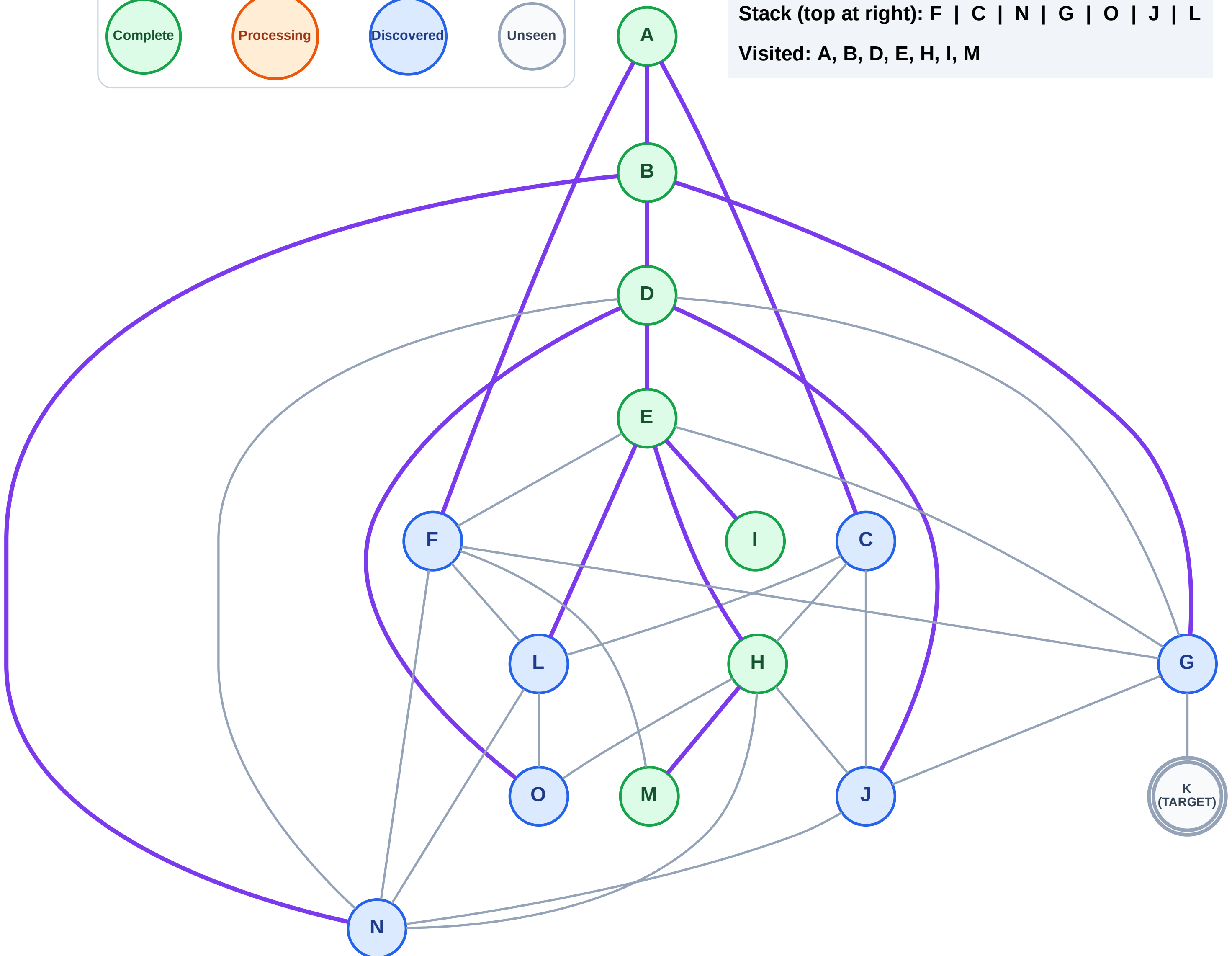
Step 15: No new neighbors from I

Legend



Stack (top at right): F | C | N | G | O | J | L

Visited: A, B, D, E, H, I, M



DFS Traversal

Step 16: Process node L

Legend

Complete

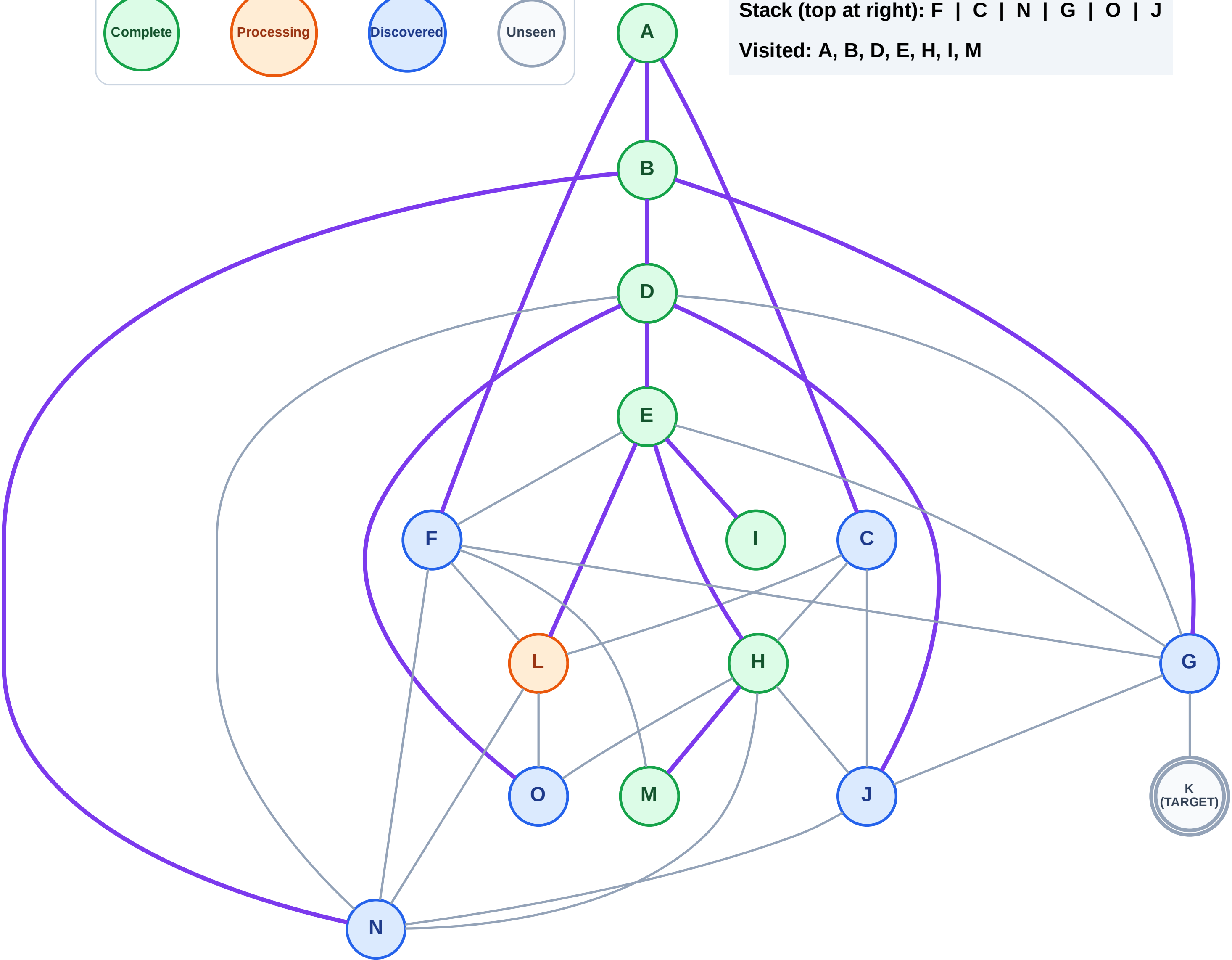
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O | J

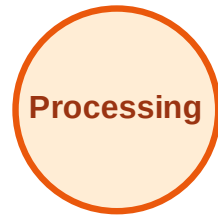
Visited: A, B, D, E, H, I, M



DFS Traversal

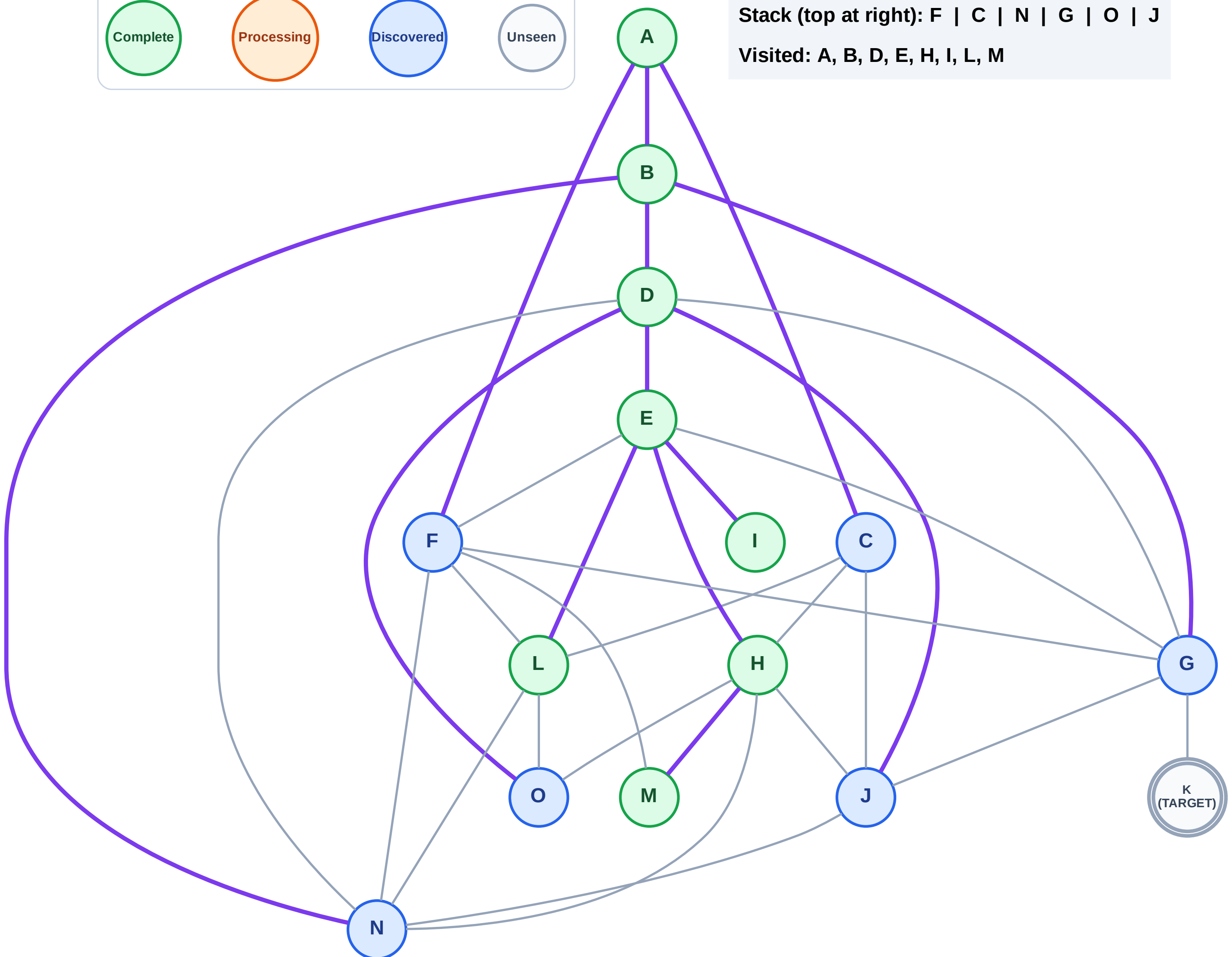
Step 17: No new neighbors from L

Legend



Stack (top at right): F | C | N | G | O | J

Visited: A, B, D, E, H, I, L, M



DFS Traversal

Step 18: Process node J

Legend

Complete

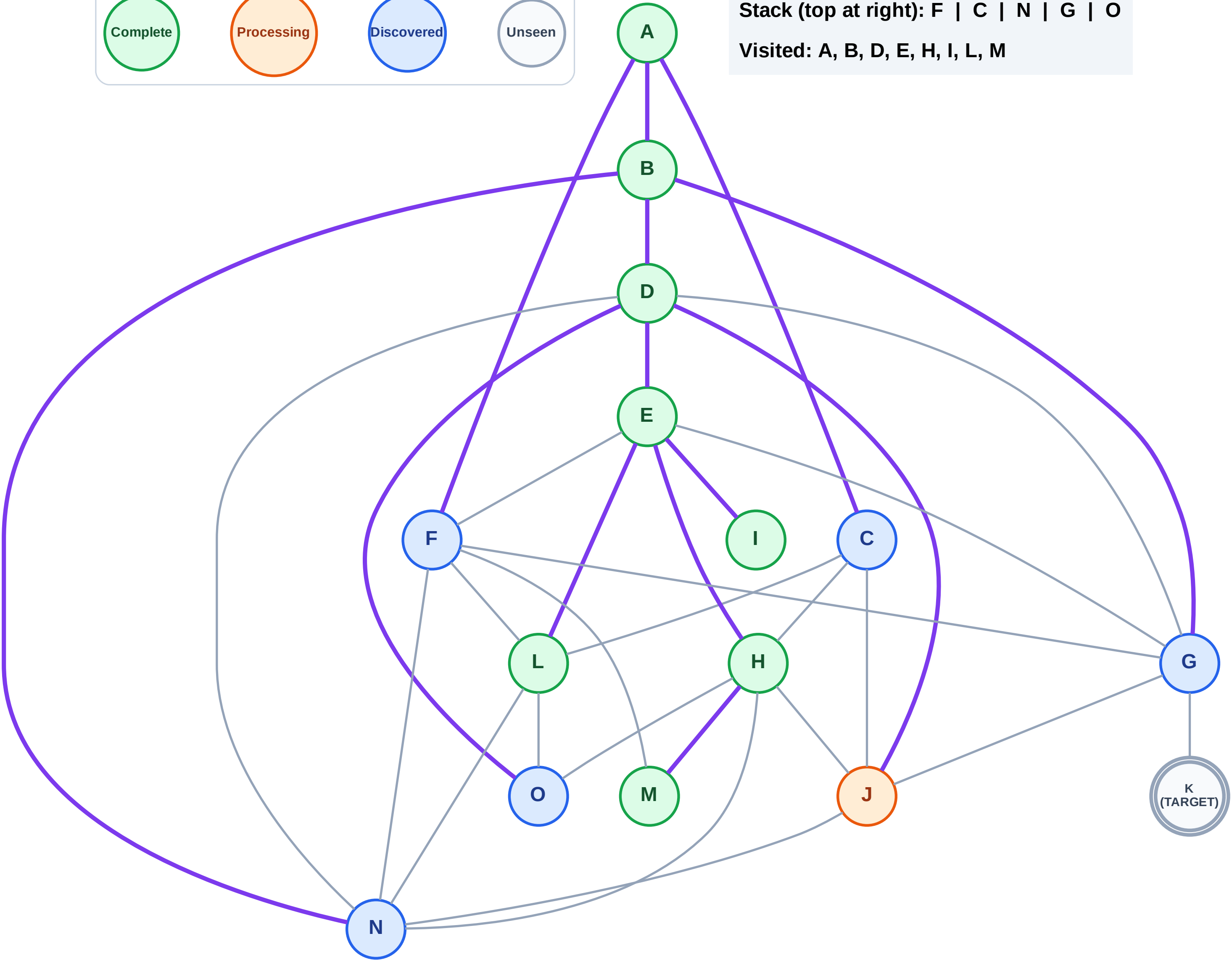
Processing

Discovered

Unseen

Stack (top at right): F | C | N | G | O

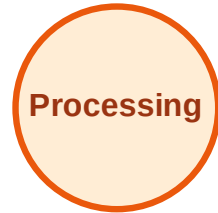
Visited: A, B, D, E, H, I, L, M



DFS Traversal

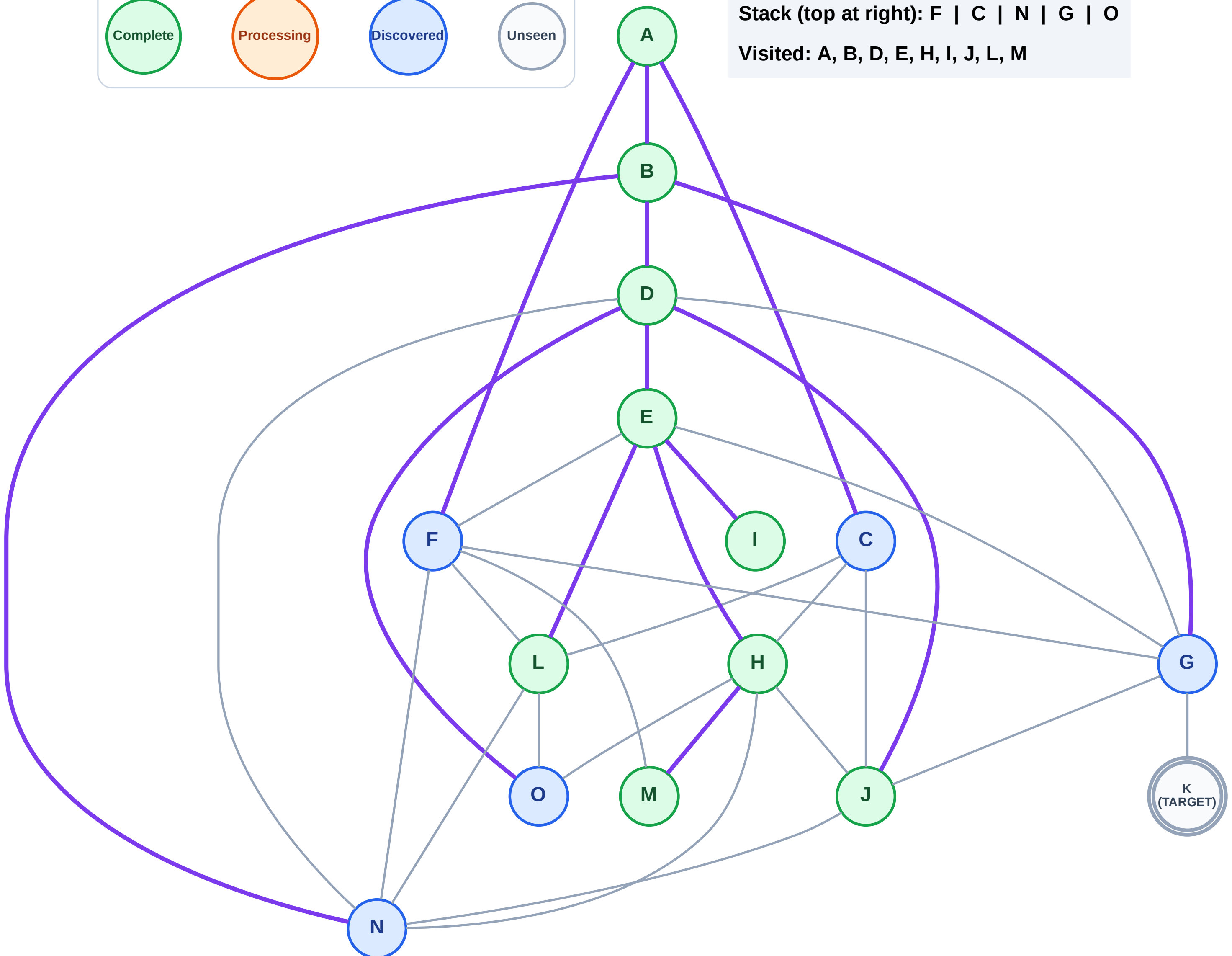
Step 19: No new neighbors from J

Legend



Stack (top at right): F | C | N | G | O

Visited: A, B, D, E, H, I, J, L, M



DFS Traversal

Step 20: Process node O

Legend

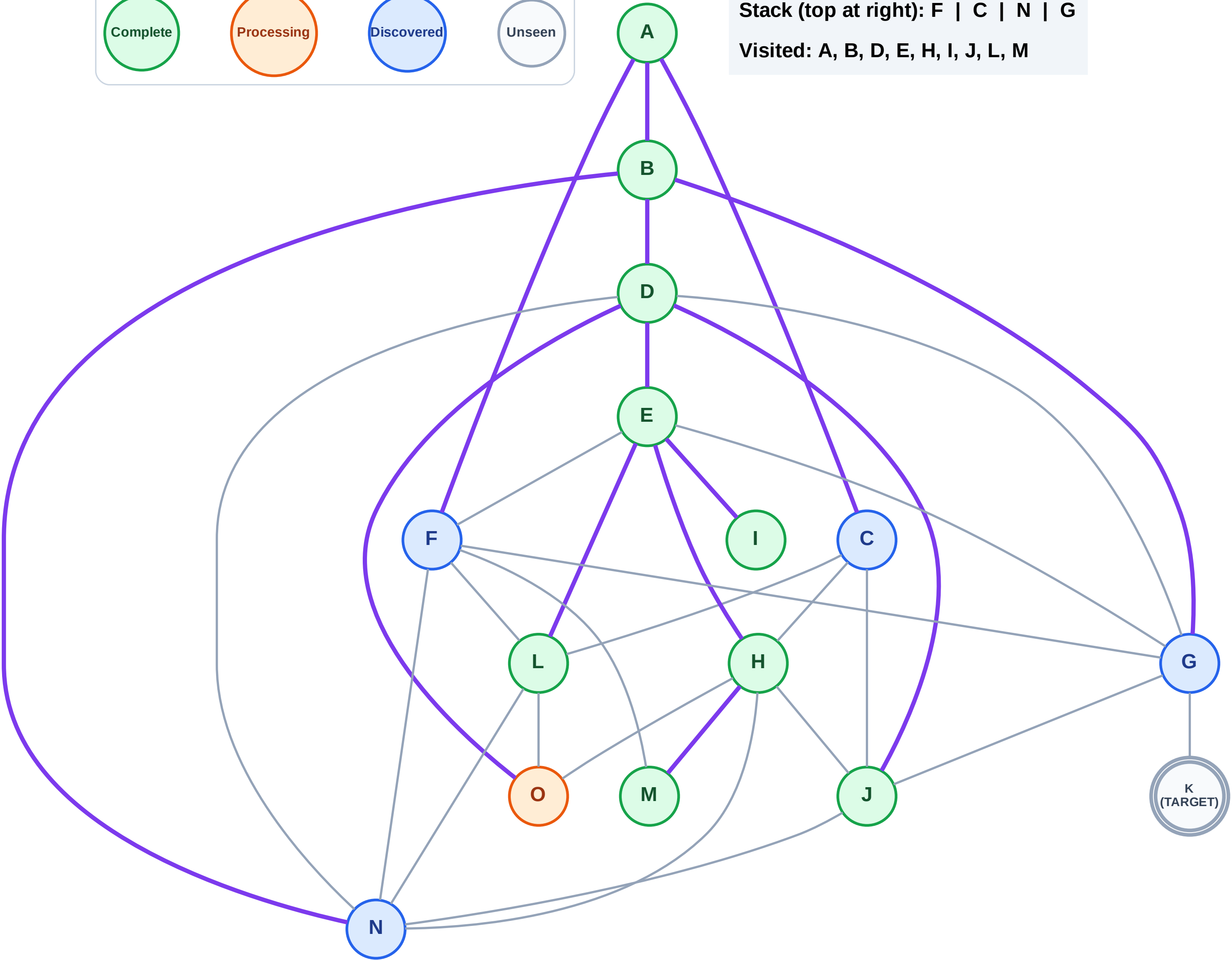
Complete

Processing

Discovered

Unseen

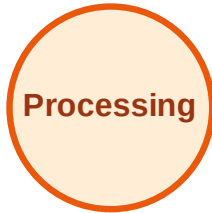
Stack (top at right): F | C | N | G
Visited: A, B, D, E, H, I, J, L, M



DFS Traversal

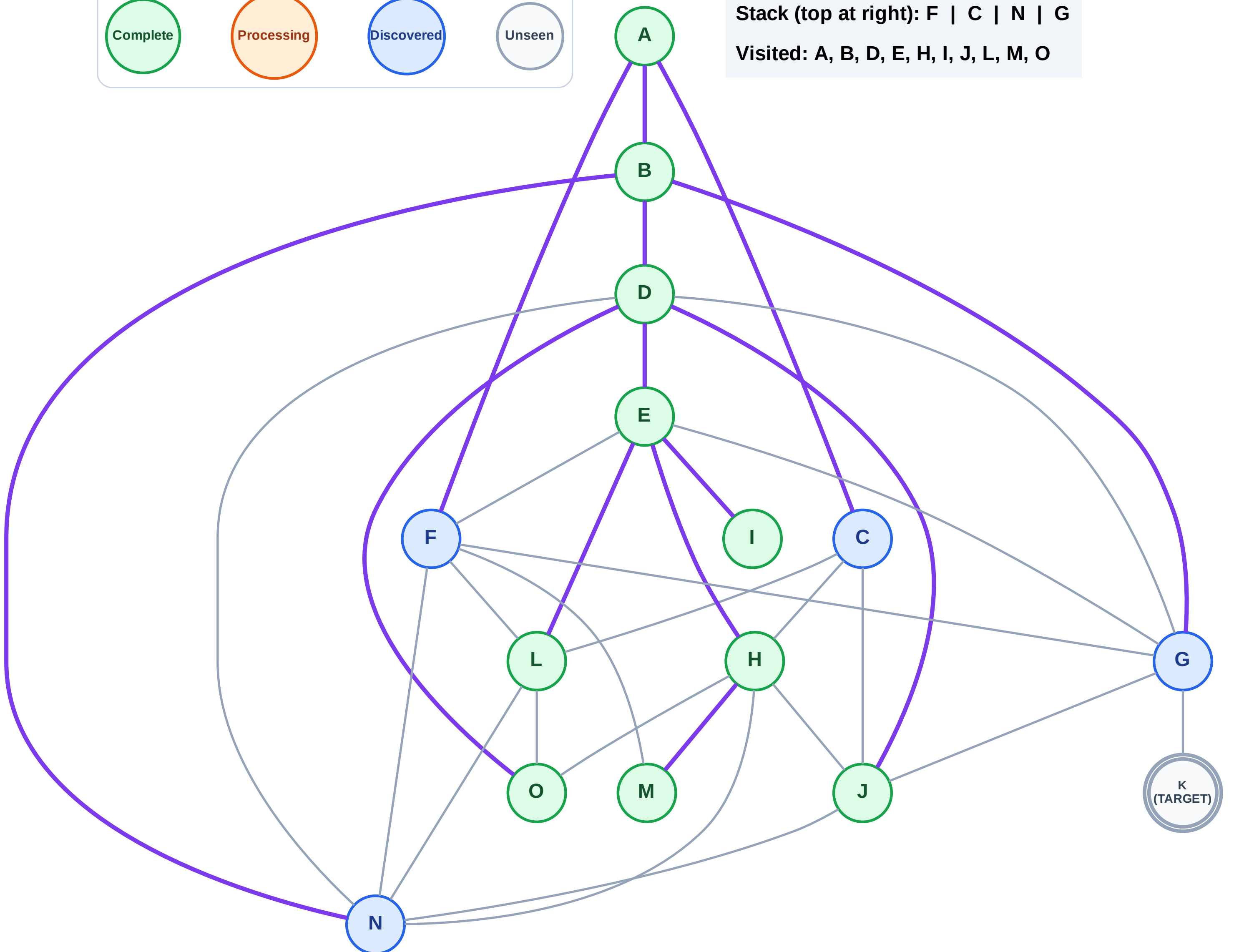
Step 21: No new neighbors from O

Legend



Stack (top at right): F | C | N | G

Visited: A, B, D, E, H, I, J, L, M, O



DFS Traversal

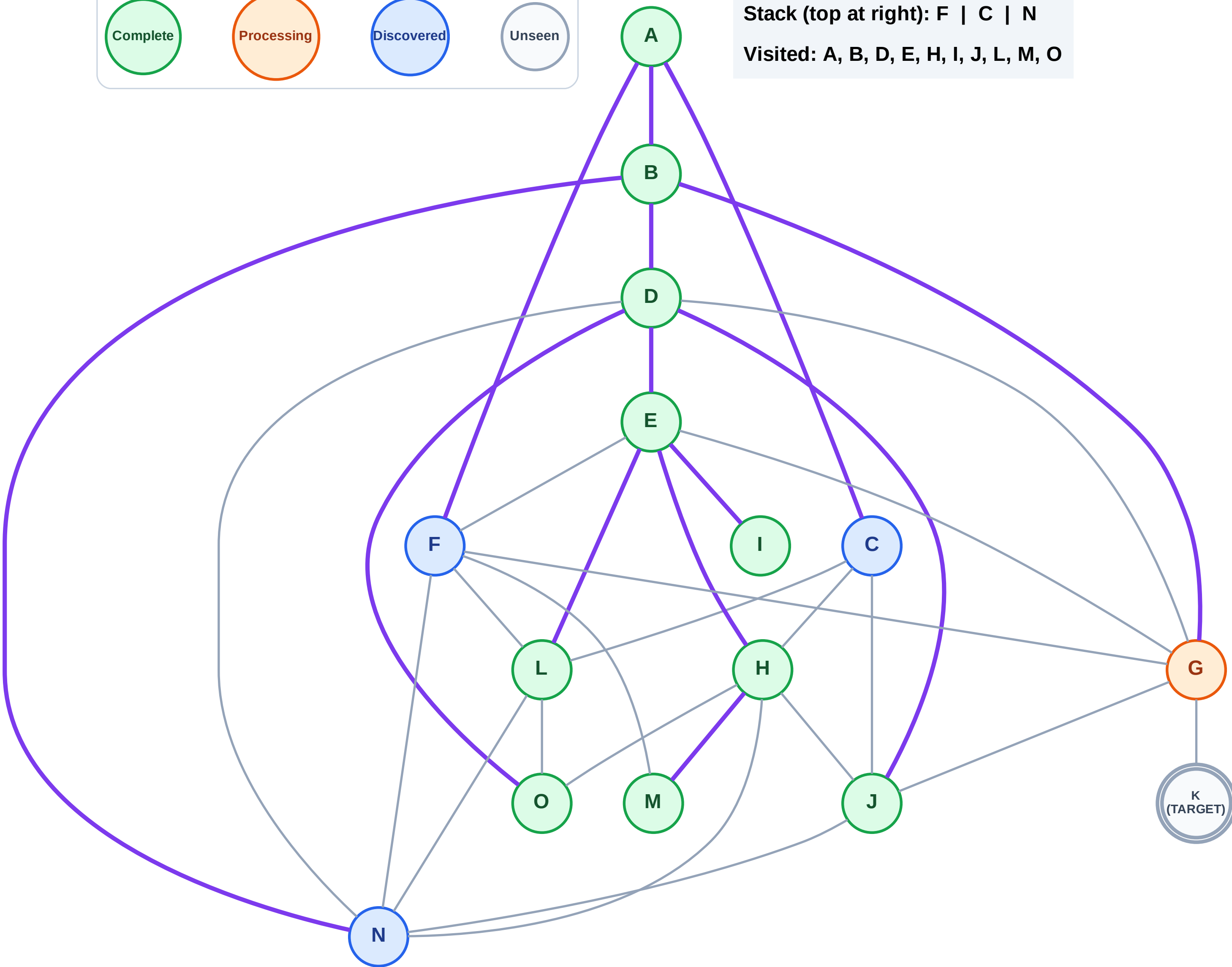
Step 22: Process node G

Legend



Stack (top at right): F | C | N

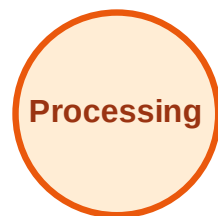
Visited: A, B, D, E, H, I, J, L, M, O



DFS Traversal

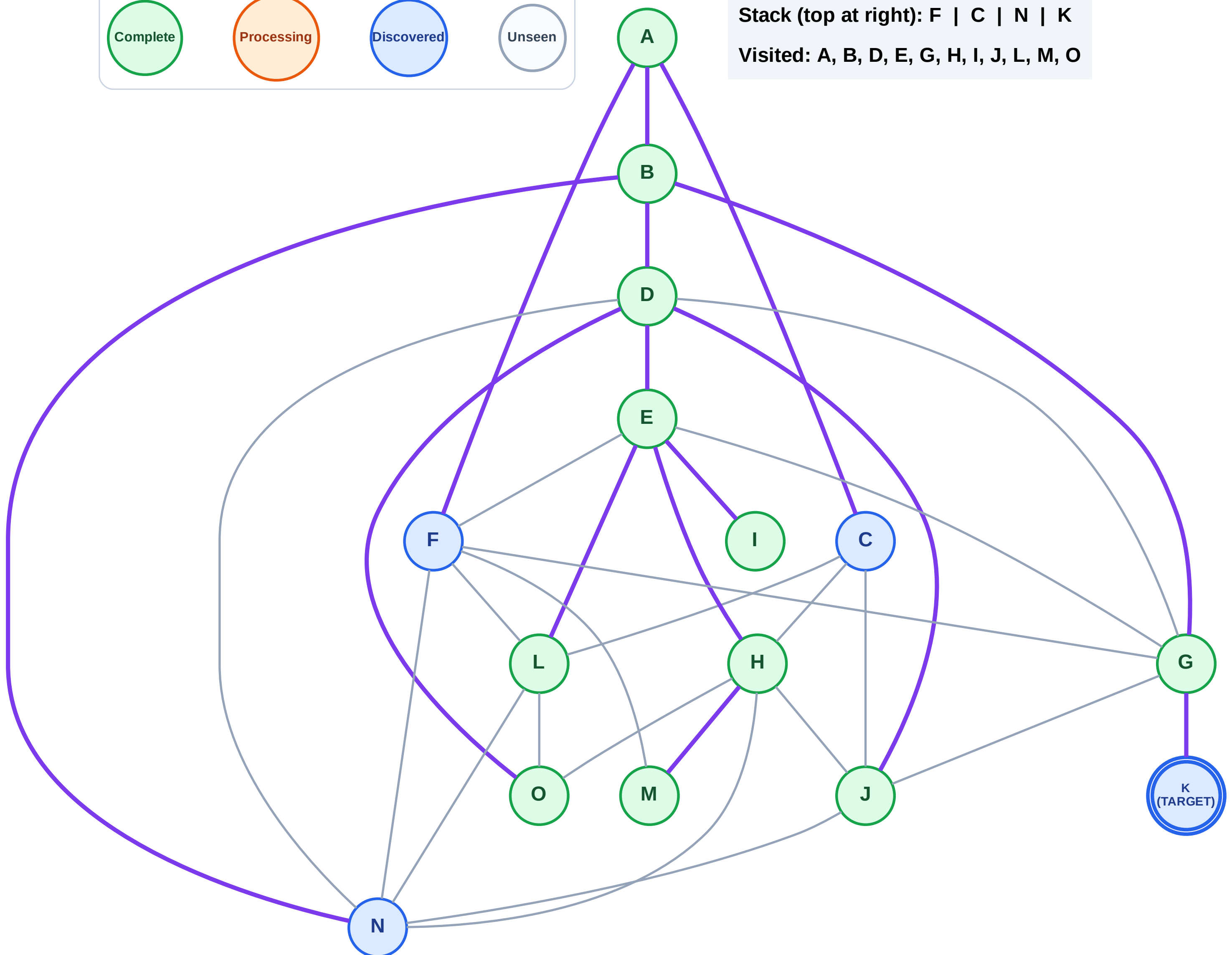
Step 23: Discover K from G

Legend



Stack (top at right): F | C | N | K

Visited: A, B, D, E, G, H, I, J, L, M, O



DFS Traversal

Step 24: Process node K

Legend

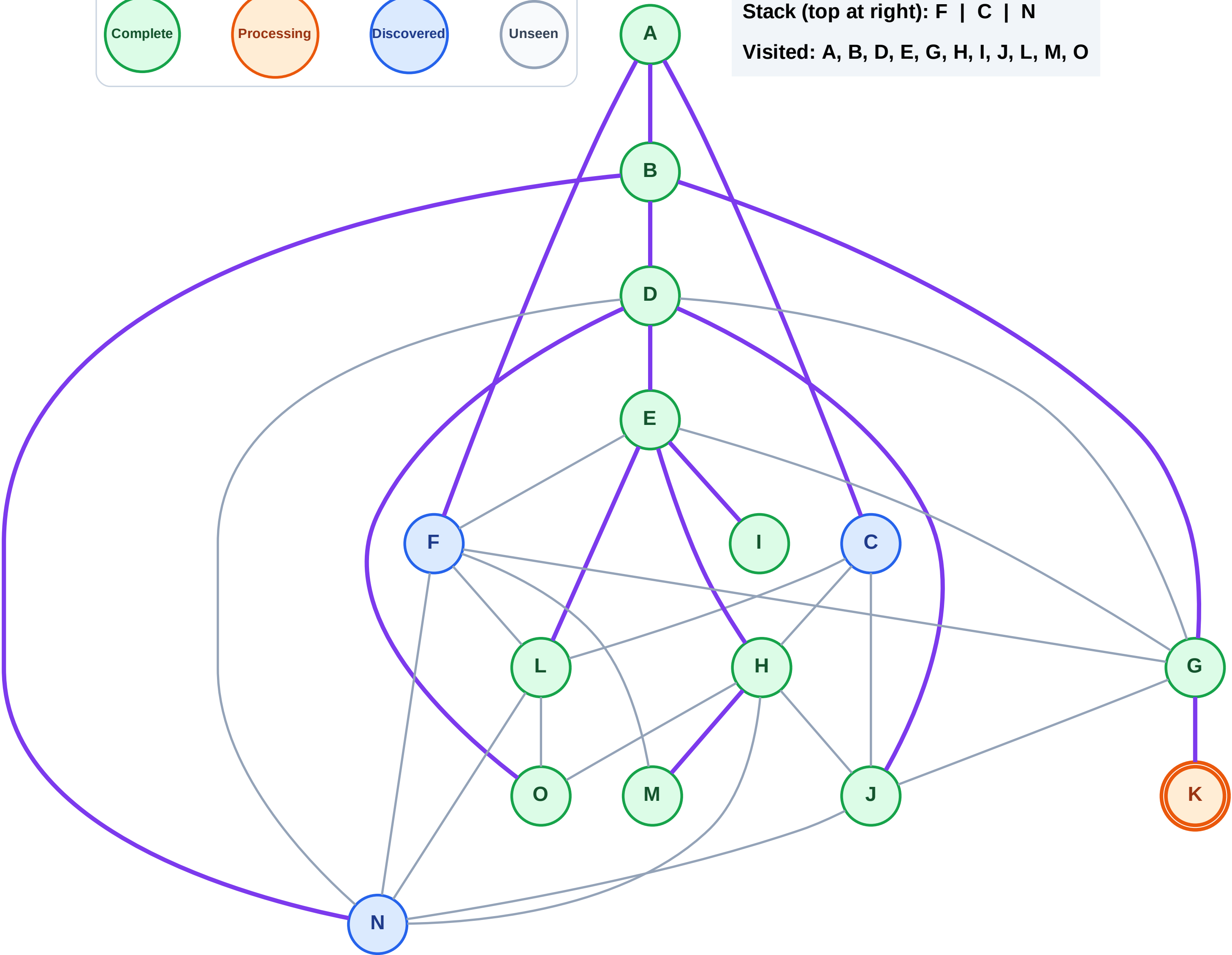
Complete

Processing

Discovered

Unseen

Stack (top at right): F | C | N
Visited: A, B, D, E, G, H, I, J, L, M, O



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (3 hops): A → B → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): F | C | N

Visited: A, B, D, E, G, H, I, J, K, L, M, O

